



MATERI HANYA DIGUNAKAN TERBATAS UNTUK PERKULIAHAN
TI5003 PEMODELAN DAN BERPIKIR SISTEM

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2. SYSTEM CONCEPT

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LEARNING OUTCOMES

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Mahasiswa dapat:

1. Memahami feedback system thinking
2. Memahami dinamika masalah dan solusinya



LEARNING OUTCOMES

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Students are able

1. to define system.
2. to understand system behaviour and the importance of emergent properties of systems.
3. to understand various classifications of systems.
4. to understand control of systems — for achieving certain desired goals of system.



OUTLINE

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1. Pervasiveness of system
2. System definition
3. System behaviour
4. Classification of system
5. Feedback loops
6. Control of systems
7. Response lags in systems



MICRO LECTURE VIDEOS

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Watch the following videos about introduction to system thinking

1. What Is A System

(https://www.youtube.com/watch?v=dTkdDx_0aRU&t=176s)



1. PERVASIVENESS OF SYSTEM

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Solar System **Political Systems**
Legal System **Digestive System**
Social Systems
Central Nervous System. **Telephone System**
Communication System
Number Systems **Information Systems**



1. PERVASIVENESS OF SYSTEM

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NATURAL SYSTEMS

- Solar System
- Biological System

ABSTRACT SYSTEMS

- Number Systems
- Information Systems

HUMAN ACTIVITY SYSTEMS

- Social System
- Business System
- Industrial System



2. DEFINITION OF SYSTEM

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Webster's 9th New Collegiate Dictionary

sys-tem n [LL systemat-, systema. fr Gk systemat-, systema. fr. synistanai to combine, fr. syn- + histanai to cause to stand – more at STAND] (1619) **1** : a regularly interacting or interdependent group of items forming a unified whole {a number ~}: as **a** (1) : a group of interacting bodies under the influence of related forces {a gravitational ~} (2) : an assemblage of substances that is in or tends to equilibrium {a thermodynamic ~} **b** (1) : a group of body organs that together perform one or more vital functions {the digestive ~} (2) : the body considered as a functional unit **c** : a group of related natural objects or forces {a river ~} **d** : a group of devices or artificial objects or an organization forming a network esp. for distributing something or serving a common purpose {a telephone ~} {a heating ~} {a highway ~} {a data processing ~} **e** : a major division of rocks usu. larger than a series and including all formed during a period or era **f** : a form of social, economic, or political organization or practice {the capitalist ~} **2** : an organized set of doctrines, ideas, or principles usu. intended to explain the arrangement or working of a systematic whole {the Newtonian ~ of mechanics} **3 a** : an organized or established procedure {the touch ~ of typing} **b** : a manner of classifying, symbolizing, or schematizing {a taxonomic ~} {the decimal ~} **4** : harmonious arrangement or pattern : ORDER {bring ~ out of confusion – Ellen Glasgow} **5** : an organized society or social situation regarded as stultifying: ESTABLISHMENT 2 – usu. used with the syn see METHOD – sys-tem-less.

The key terms:

- Interacting, interdependent, and forming a unified whole.
- The key emphasis is on **systemic** (not systematic) or **organized** i.e. pertaining to systems, using systems ideas, or viewing things in terms of their role in a system.



2. DEFINITION OF SYSTEM

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Out-there and inside-us view of systems

- **Out-there view:** For well-defined system; The system is seen as existing independently of the observer.
- **Inside-us view:** The system is not seen as existing independently of the observer; it is not out there; it has become a mental construct, personal to the observer!

Systems as a human conceptualization

- Systems are seen as human conceptualizations
- It is the inside-us view of system.



2. DEFINITION OF SYSTEM

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Formal Definition

- A system is an *organized assembly* of components.
- The system *does* something.
- Each component *contributes* towards the behavior of the system and is affected by being in the system.
- Groups of components within the system may form *subsystem*.
- The system has an outside- an *environment*.
- The system has been identified by someone as of *special interest*.



2. DEFINITION OF SYSTEM

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Out-there view

- Pandangan yang melihat sistem sebagai benda fisik dan abstrak yang membentuk kumpulan, hubungan dan yang aktivitas sistem.
- Sistem dilihat sebagai sesuatu yang absolut – ada atau akan ada di luar sana.
- Sistem dilihat sebagai sesuatu yang independen dari pengamat.
- Contoh: **sistem matahari** (*solar system*), semua memiliki definisi yang sama ↔ **sistem tenaga listrik** atau **sistem informasi komputer**.

Inside-us view

- Definisi sistem sangat tergantung pada sudut pandang orang dan tujuan dalam mendefinisikan sistem tersebut.
- Sistem sebagai konseptualisasi manusia.
- Contoh: **sistem tenaga listrik** dapat mencakup struktur biaya listrik (tergantung dari interest dan tujuan pendefinisian sistem) → masalah yang akan ditangani.



2. DEFINITION OF SYSTEM

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Example:



Grandfather: views estuary (jeram) a beautiful place to share the wonders of nature with his grandchild.

Jogger (crossing their path): may be hardly aware of anything more than a few feet away from the path.

**Estuary
IS NOT a system**

Biology student: studying the estuary will see it as an ecological system, where plants, insects, and all sorts of aquatic life forms interact with each other and are affected by the tides

**Estuary
IS a system**

Engineer (engineer working for the local catchment authority) will see estuary as a system, in fact, a **subsystem** of a larger water drainage system under her management.



3. SYSTEM

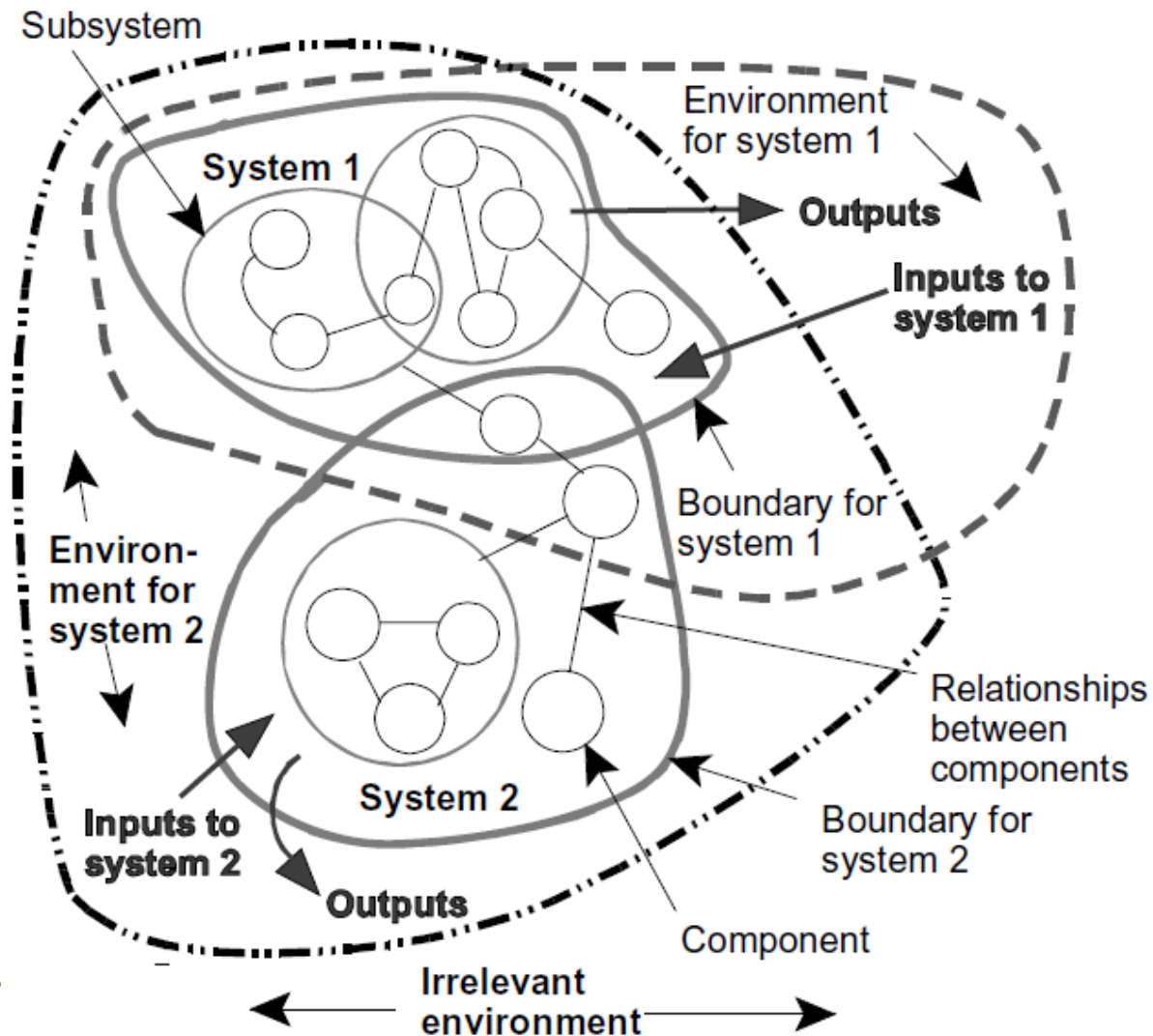
Aspek Sistem:

- Lingkungan: *Inner-system boundary-outer.*
- *Open system vs. closed system.*
- **Kandungan sistem:**
 - Seluruh elemen yang tercakup;
 - *Tanpa hubungan antar elemen.*
- **Struktur system:** mencakup pola hubungan antar elemen.
- **Subsystem:** kumpulan parsial dari elemen-elemen suatu system.
- Beberapa aspek system: teknikal, ekonomik, sosio-psikologis, dsb. → dapat saling berhubungan (interrelasi) → *a multi-aspect system.*
- Dalam suatu sistem multi aspek, analisis-analisis masalah parsial perlu digabungkan menjadi analisis masalah secara terpadu (integral).



3. SYSTEM

Aspek Sistem:



Source: Daellenbach, H. G. (2005).



3. SYSTEM

Elemen dan Hubungan antar Elemen:

- Elemen suatu system dicirikan dengan fitur tertentu, seperti: fisikal, geometrical, estetika, sosio-psikologgical atau ekonomi.
- Jika terdapat hubungan antar elemen, maka jika karakteristik suatu elemen berubah, maka karakteristik elemen yang lain akan berubah dengan cara serupa.
- Hubungan antar elemen dapat dalam aspek teknikal, sosio, ekonomik, sosio-psikologikal.

Elemen Krusial Sistem:

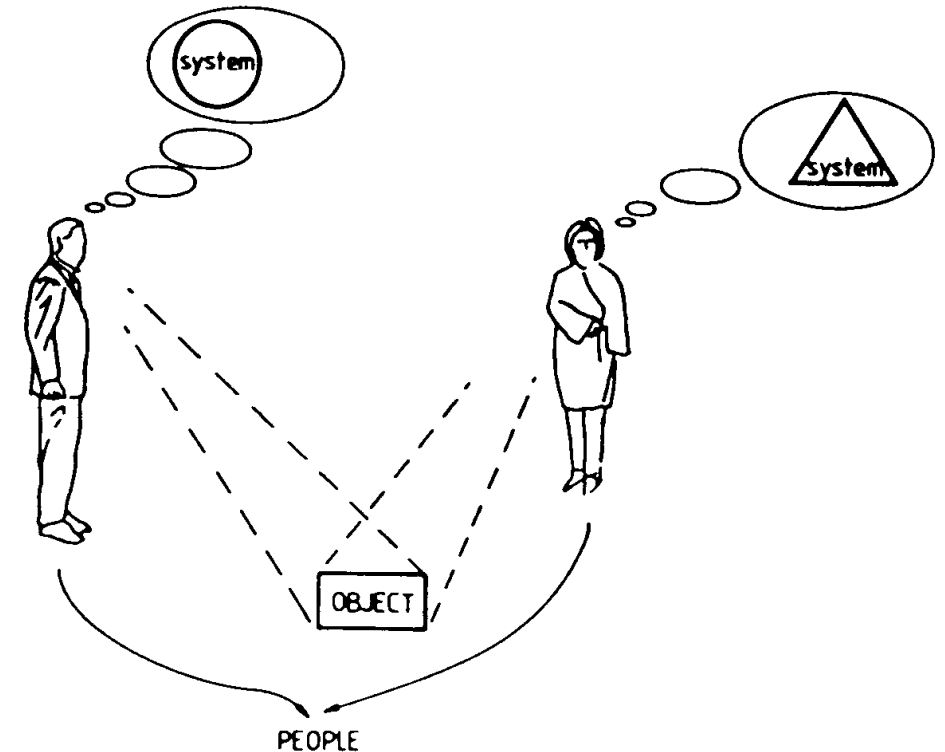
- Komponen
- Hubungan
- Perilaku (atau aktivitas, atau proses transformasi);
- Lingkungan;
- Input dari lingkungan dan output ke lingkungan;
- Minat tertentu dari pengamat.



3. SYSTEM

Subyektivitas:

- Sudut pandang (world view of observer) atau faktor personal;
- Efek dari pengetahuan sebelumnya;
- Definisi sistem adalah subyektif:
 - *cannot be labelled “right” or “valid” & “wrong” or “invalid”;*
 - *valid for the person making it;*
 - *dependent on the aim and purpose for building it;*
- Sistem merupakan konseptualisasi manusia;
- *Systems as a human conceptualization.*
- Hanya pengamat manusia yang dapat melihat sesuatu sebagai sistem.





3. SYSTEM

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Subyektivitas:

*A System is a Way of
Looking at the World*





3. SYSTEM

A System Is a Way of Looking at the World



Do you see three faces?



Find the two faces in the flowers



3. SYSTEM

Subyektivitas:

- Half full – half empty glass.....

Which one are you?



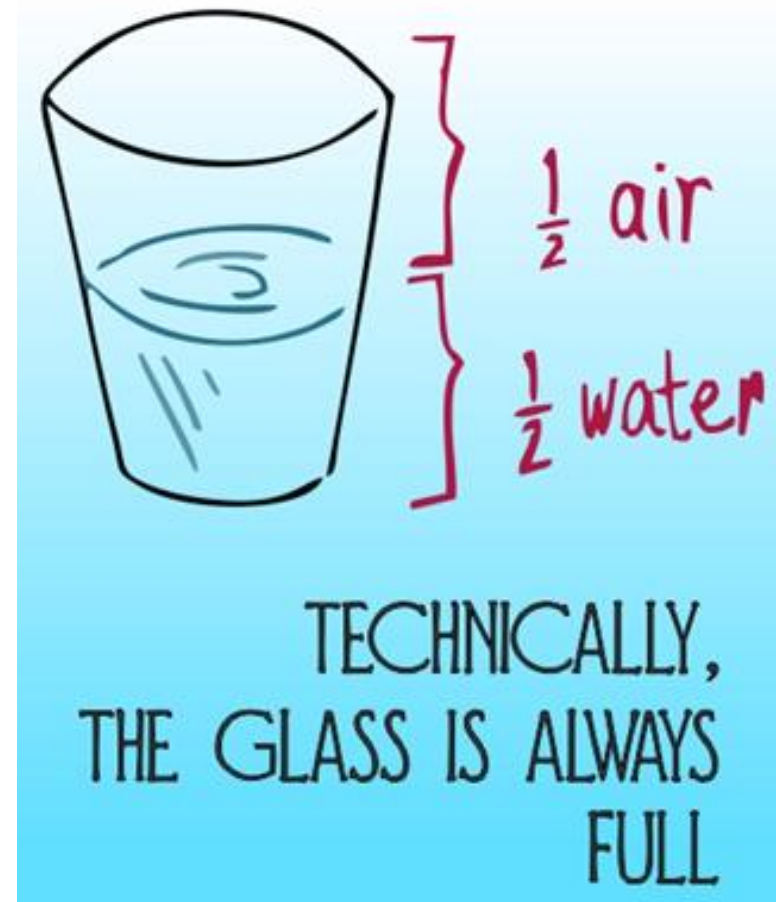


3. SYSTEM

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Subyektivitas:

- **Half full –
half empty glass.....**

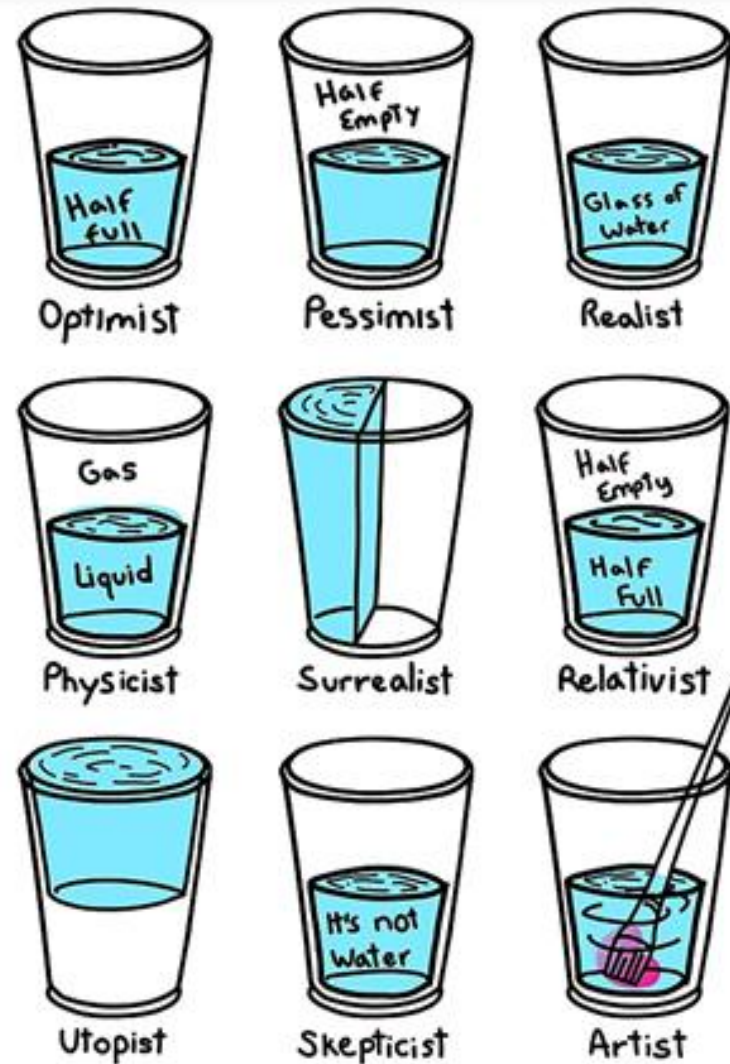




3. SYSTEM

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Subyektivitas:





3. SYSTEM

Sistem dalam Konteks Pengambilan Keputusan:

A System is described by:

- **Observer:** Who is interested in the system? Decision maker.
- **Purpose:** Why define the system? Improve, output of interest.
- **Environment:** Outside the system. Define the system boundary.
- **Inputs:** Affect the system but are not affected by it. Can be controllable or uncontrollable. Decision variables/parameters.
- **Outputs:** Are affected by the system. These are of interest to the observer. Include measure of success
- **Components:** Both affect and are affected by the system.
- **Relationship/transformation process:** Between system inputs, outputs and components.



3. SYSTEM

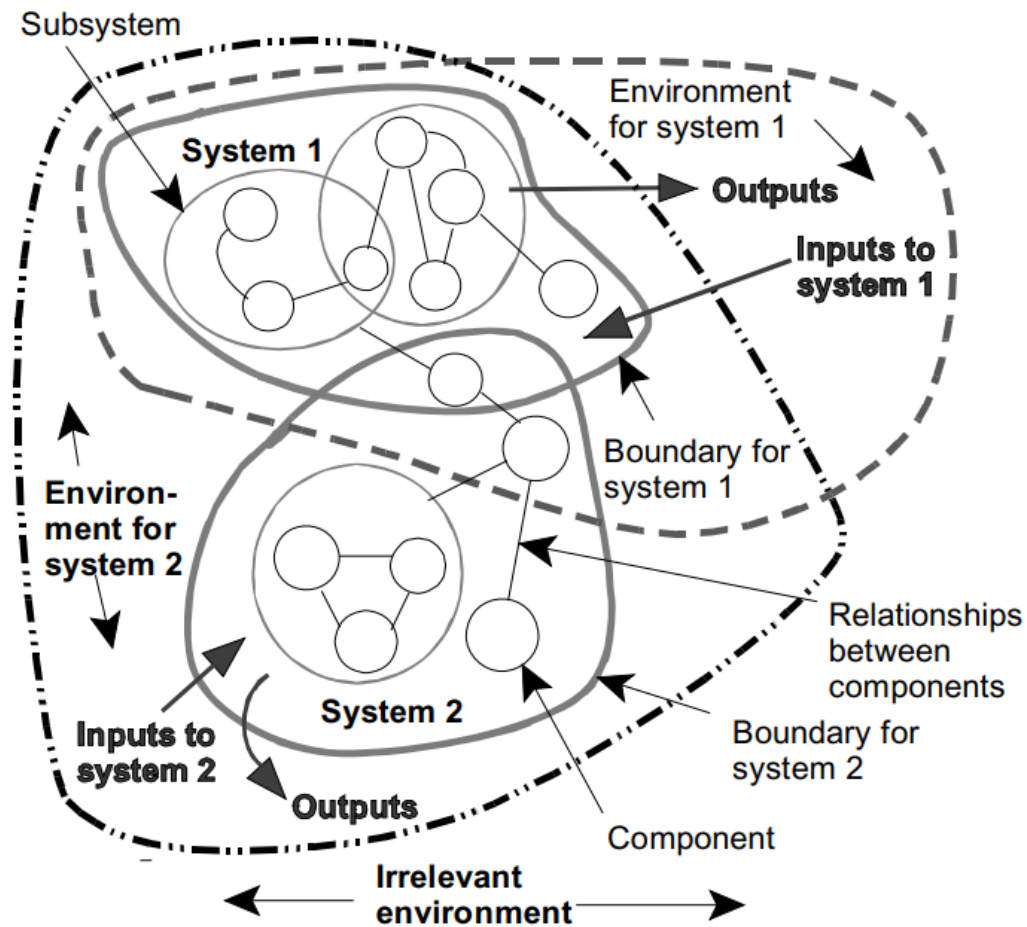
Hirarki Sistem:

- Melihat sesuatu sebagai sistem berdampak pada:
 - Aspek apa yang harus dicakup sebagai bagian dari sistem, dan
 - Aspek apa yang lebih tepat ditempatkan dalam lingkungan (*wider system*).
- Suatu sistem dapat menjadi bagian dari suatu sistem yang lebih besar atau suatu sistem dapat didekomposisikan menjadi sistem yang lebih kecil.
- Penempatan (*nesting*) suatu sistem dalam sistem disebut sebagai hirarki suatu sistem.
- Hirarki sistem menentukan batasan sistem.
- Sistem yang melingkupi menjadi lingkungan dari sistem yang dilingkupi, dan dapat mengontrol sistem yang dilingkupi tersebut.



3. SYSTEM

Environment, systems, subsystems, and components:

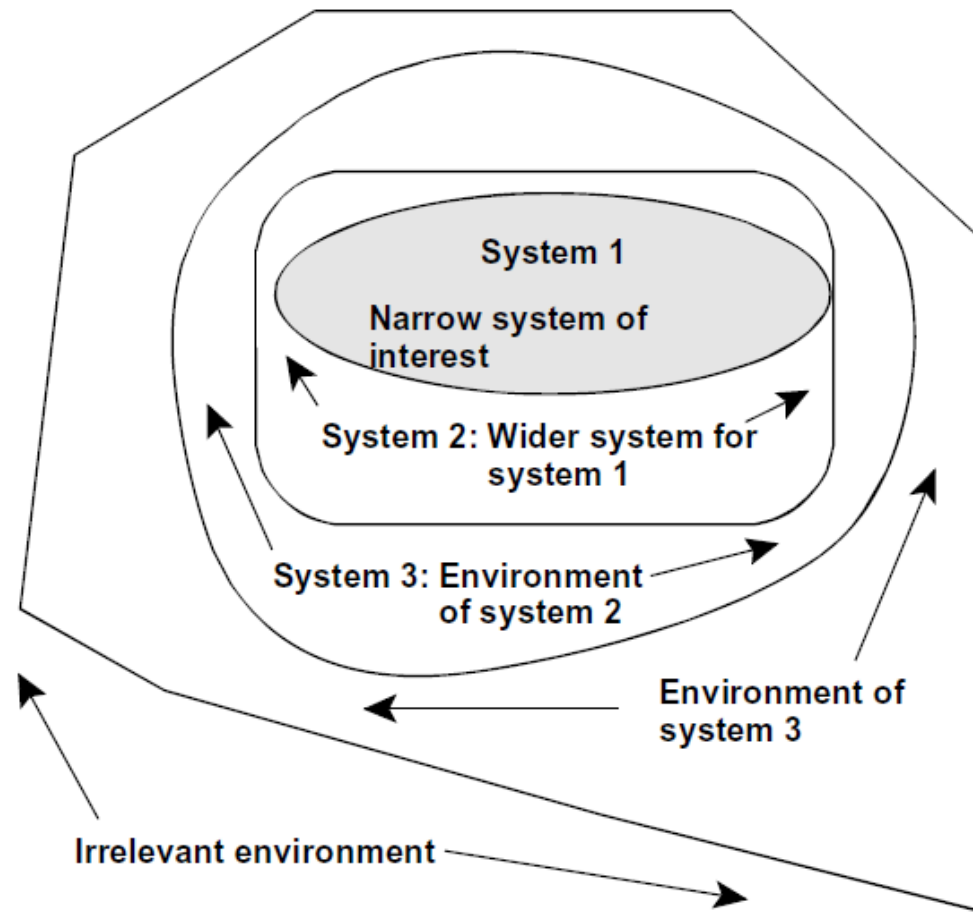


Source: Daellenbach, H. G. (2006)



3. SYSTEM

Hirarki Sistem:



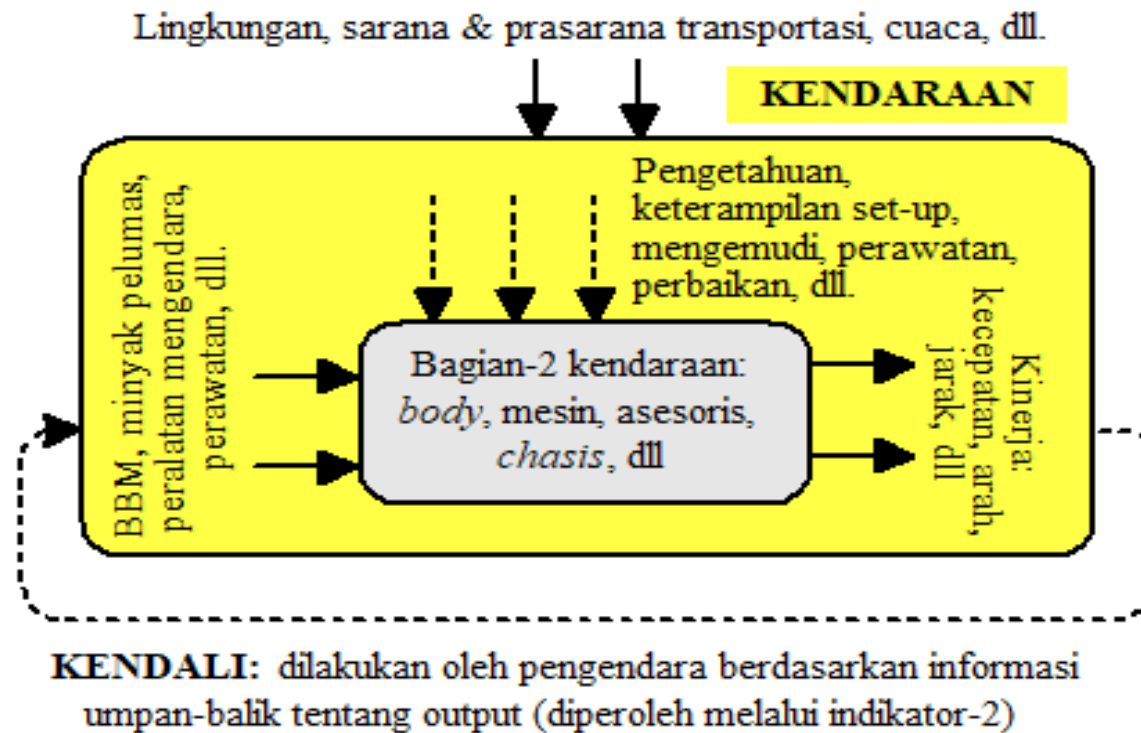
Source: Daellenbach, H. G. (2005).



3. SYSTEM

Hirarki Sistem: Sistem dan meta sistem (Sistem Fisik)

Hirarki 1: Kendaraan



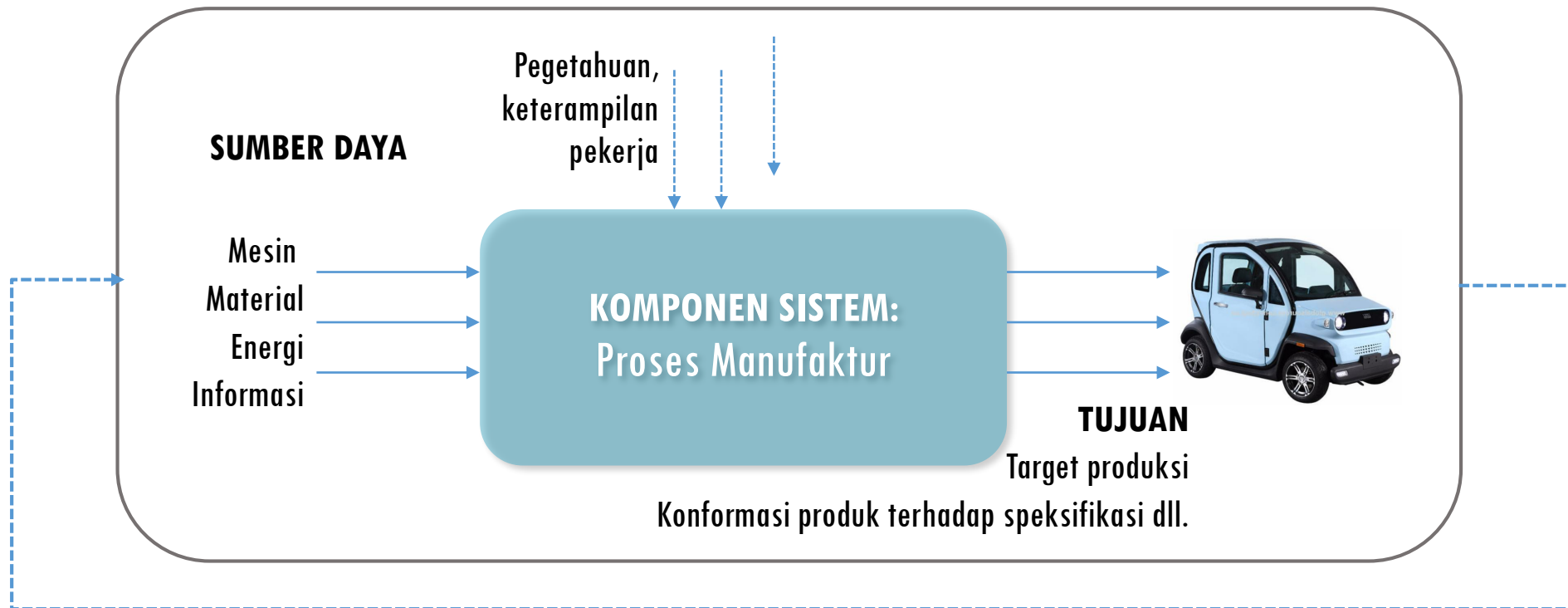


3. SYSTEM

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Contoh sistem & meta-sistem teknologi (2 hirarki): KENDARAAN & PABRIK KENDARAAN

LINGKUNGAN: Bagian lain perusahaan



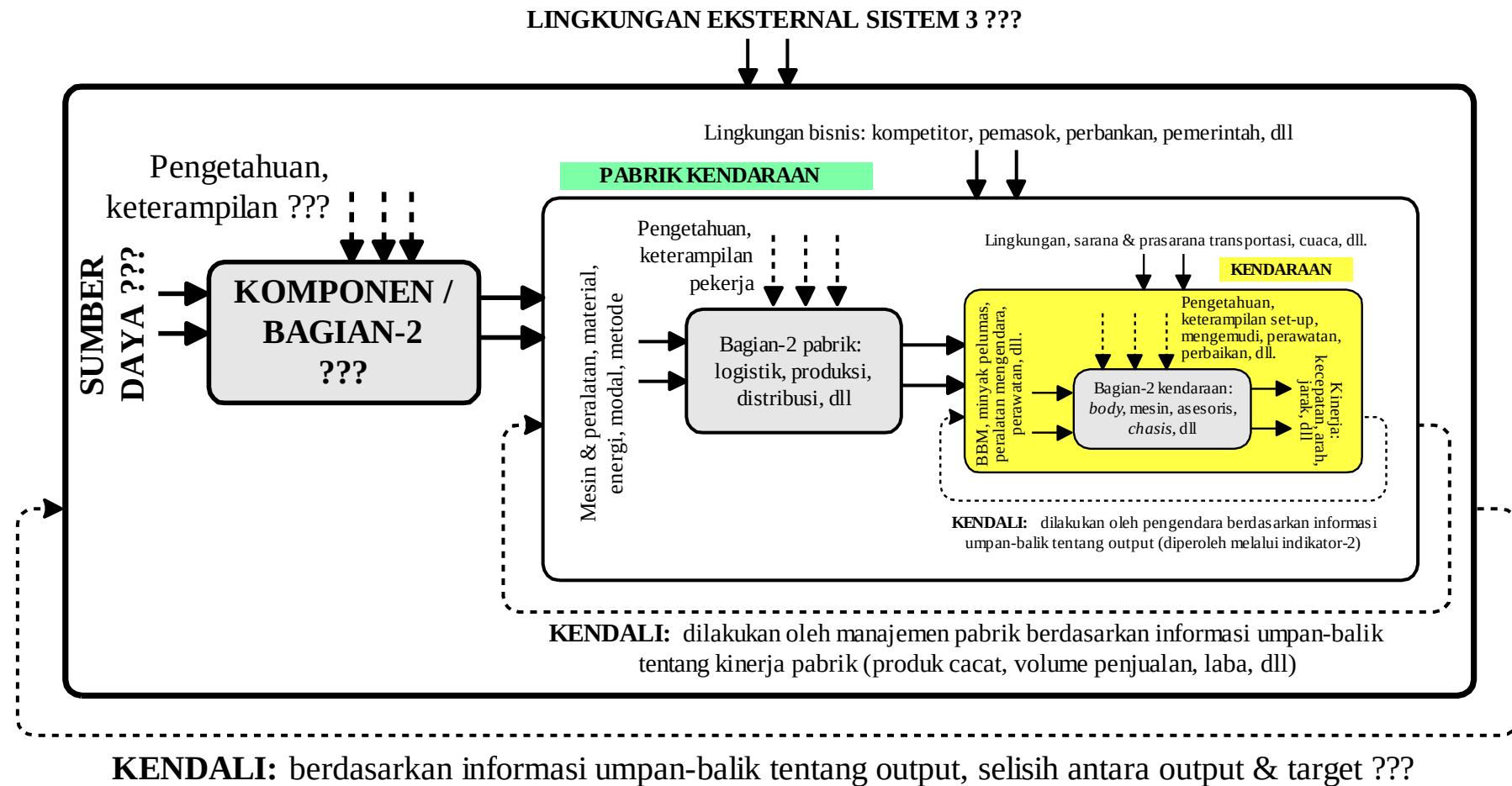
Kendali: dilakukan oleh **MANAJEMEN PABRIK** berdasarkan informasi umpan-balik tentang kinerja pabrik (produk cacat, efisiensi, produktivitas, dll)



3. SYSTEM

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Ilustrasi Sub-sistem, Sistem & Meta Sistem (3 hirarki):





4. SYSTEM BEHAVIOUR

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Status Sistem (System state)

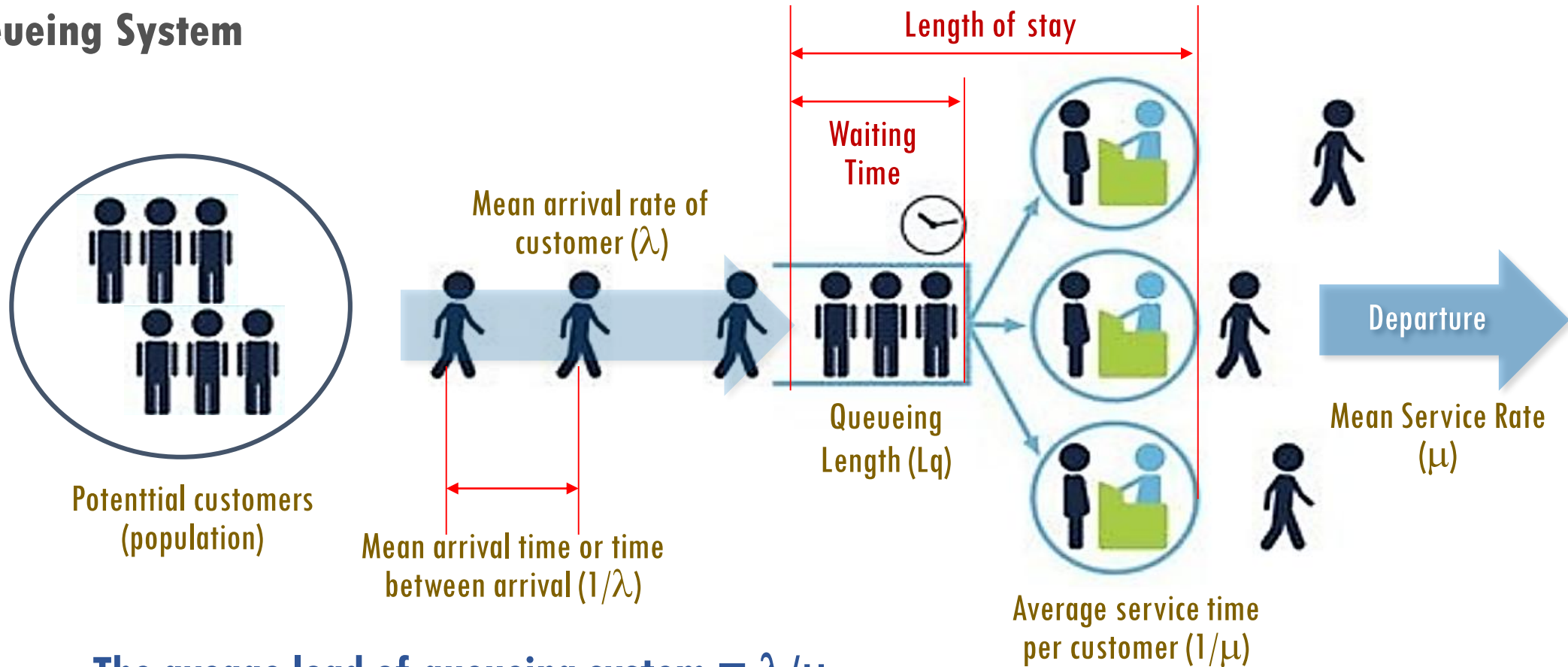
- **Perilaku sistem** merupakan obyek yang menjadi perhatian orang yang mengkaji tentang sistem.
- Perilaku sistem direpresentasikan oleh karakteristik atau atribut dari komponen sistem yang disebut sebagai **variabel status** (*state variable of system*) yang **memiliki nilai dan berubah seiring waktu**.
- Variabel status dapat memiliki **nilai numerik** (seperti kecepatan, koordinat geografis kendaraan) atau nilai kategorial (seperti kondisi jalan “tertutup”, “terbuka”).
- Status sistem dinyatakan oleh **nilai-nilai dari seluruh variabel status pada titik waktu tertentu**.
- **Perilaku sistem** ditunjukkan oleh pola **perubahan nilai dari variabel-variabel status sistem**.
- Perilaku sistem (pola perubahan nilai variabel status) digunakan sebagai dasar **evaluasi kinerja sistem**.



4. SYSTEM BEHAVIOUR

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Queueing System



The average load of queueing system = λ/μ
State variable: Waiting time or Length of stay

EXCEL

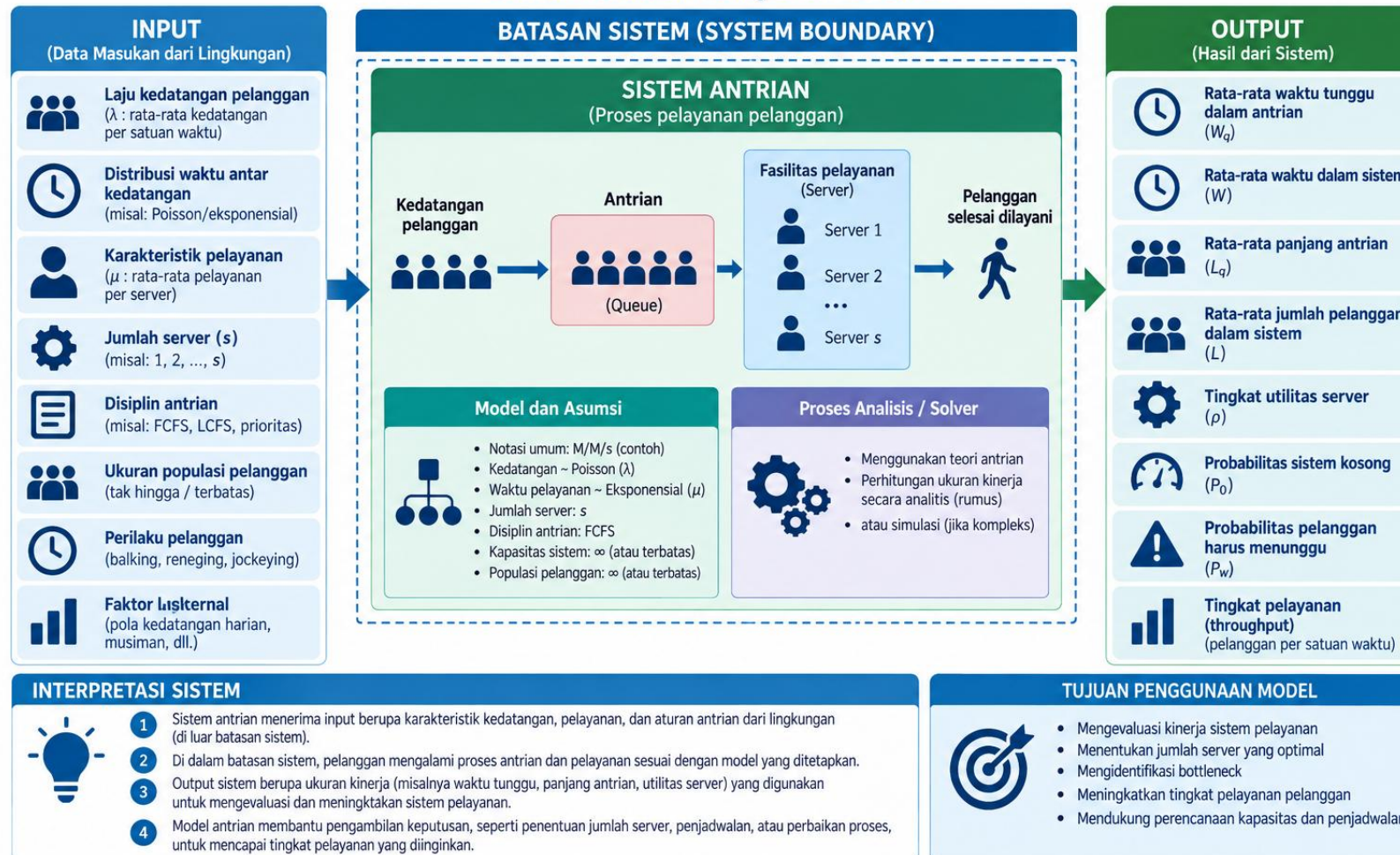


4. SYSTEM BEHAVIOUR

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Queueing System (Sistem Fisik)

Deskripsi Sistem untuk Model Antrian Contoh: Sistem Pelayanan di Loker



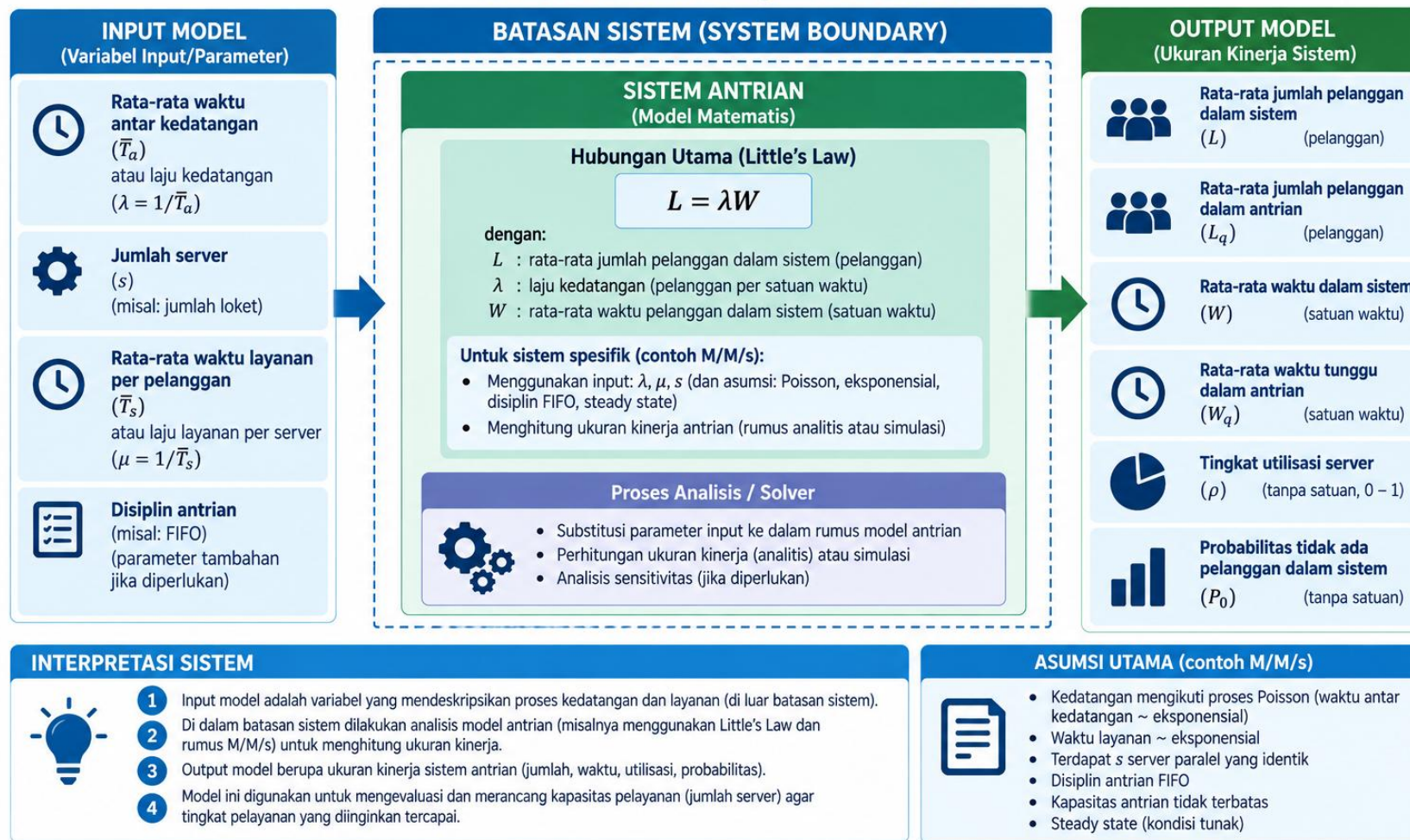


4. SYSTEM BEHAVIOUR

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Queueing System (Sistem Relevan)

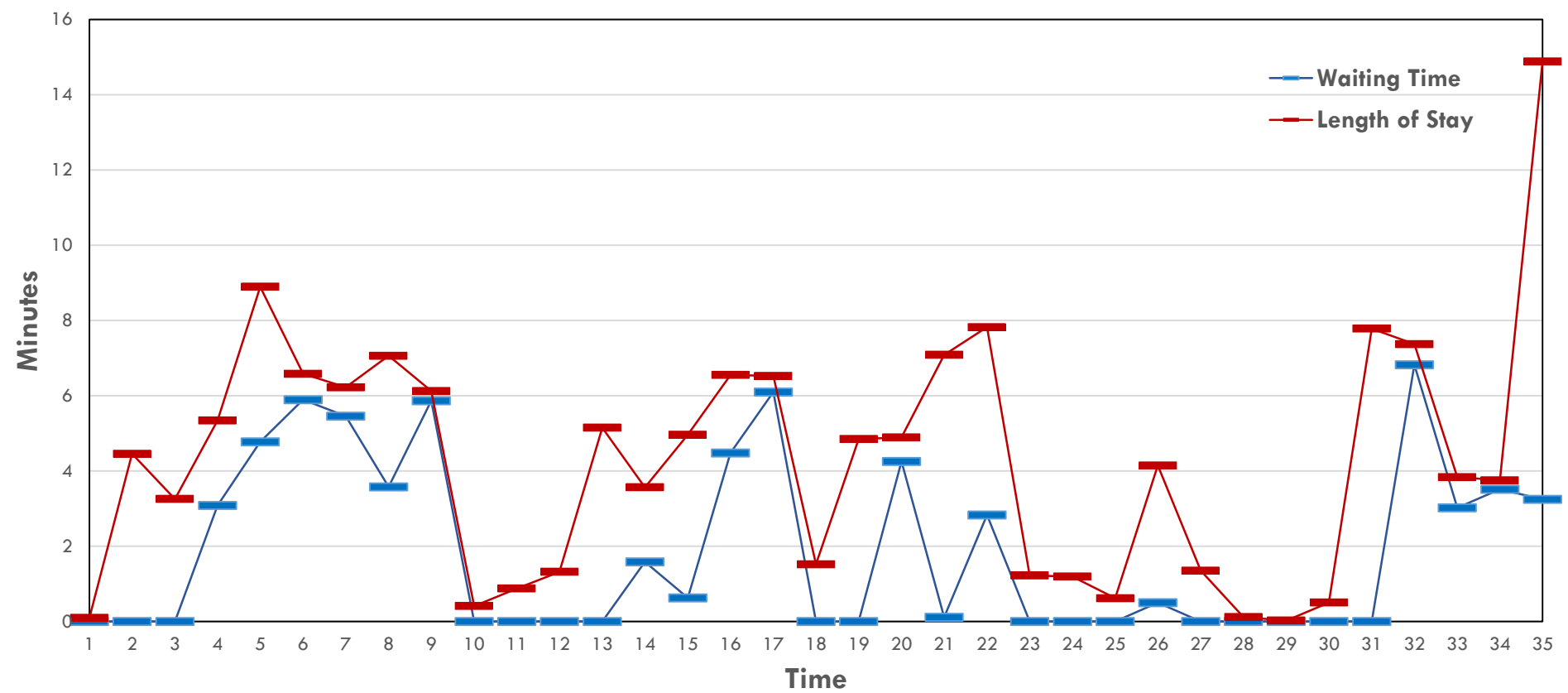
Deskripsi Sistem untuk Model Antrian (Little's Law) Contoh: Sistem Pelayanan di Loket





4. SYSTEM BEHAVIOUR

State of the Queueing System





4. SYSTEM BEHAVIOUR

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Emergent properties

- Perilaku yang ditunjukkan oleh sistem yang muncul akibat interaksi antar komponennya, tetapi tidak ditunjukkan oleh perilaku dari komponen-komponennya secara individu.
- Contoh:
Kepadatan lalu lintas pada suatu titik tertentu merupakan akibat interaksi antar individu kendaraan.



5. CLASSIFICATION OF SYSTEM

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- Discrete Systems vs Continuous Systems (perubahan status sistem secara diskrit/kontinyu)
 - **A discrete system** changes its state at discrete points in time. Between these times, the state of the system remains unchanged.
 - **A continuous system** changes its state continuously over time.
- Deterministic Systems (perilaku terprediksi) vs Stochastic Systems (perilaku dipengaruhi oleh input random atau stokastik)
 - **Deterministic System:** the behavior of the system is **predictable**.
 - **Stochastic System:** the behavior of a system is **not completely predictable**.
 - **Steady state** of a probabilistic system: a state of equilibrium of state variables of probabilistic system in the long run.
 - **Transient state** of probabilistica system: a state of state variables of probabilistic system that is influenced by initial values of its state variables.



5. CLASSIFICATION OF SYSTEM

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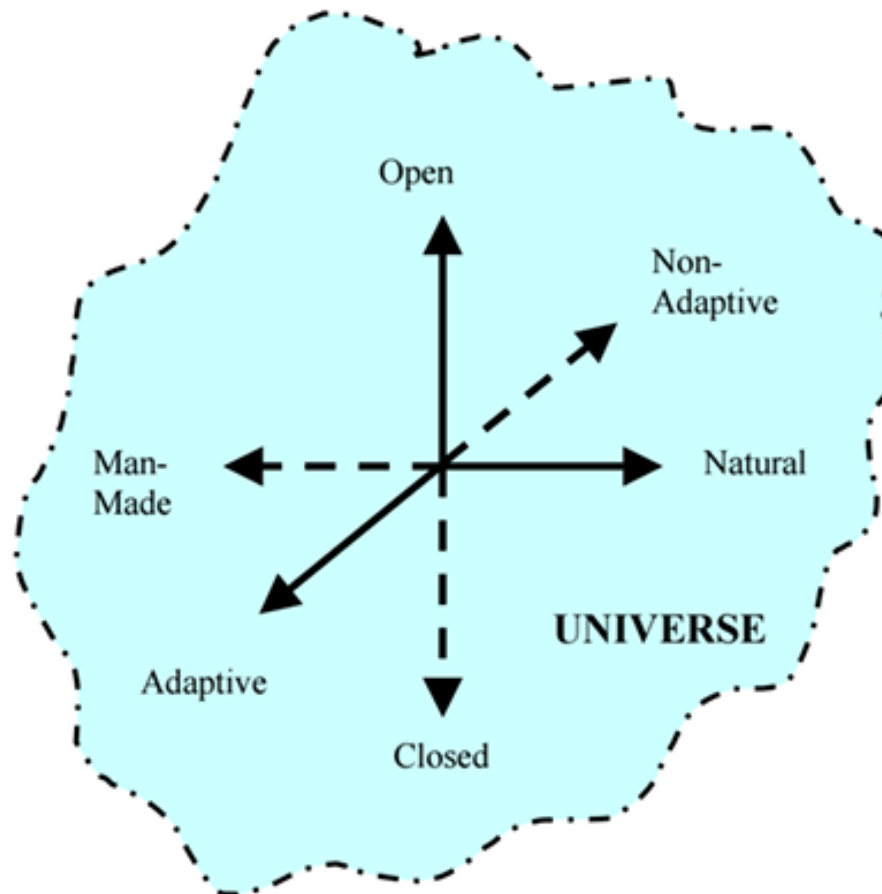
- Closed Systems (tanpa interaksi dengan lingkungan I/O) vs Open Systems (memiliki interaksi dengan lingkungan)
 - **A closed systems** is one that does not receive anything from its environment, nor does it give out anything to its environment.
 - **A open systems** An open system is one that interacts with the environment, by receiving inputs from it and providing outputs to it.



5. CLASSIFICATION OF SYSTEM

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Attributes of System Type



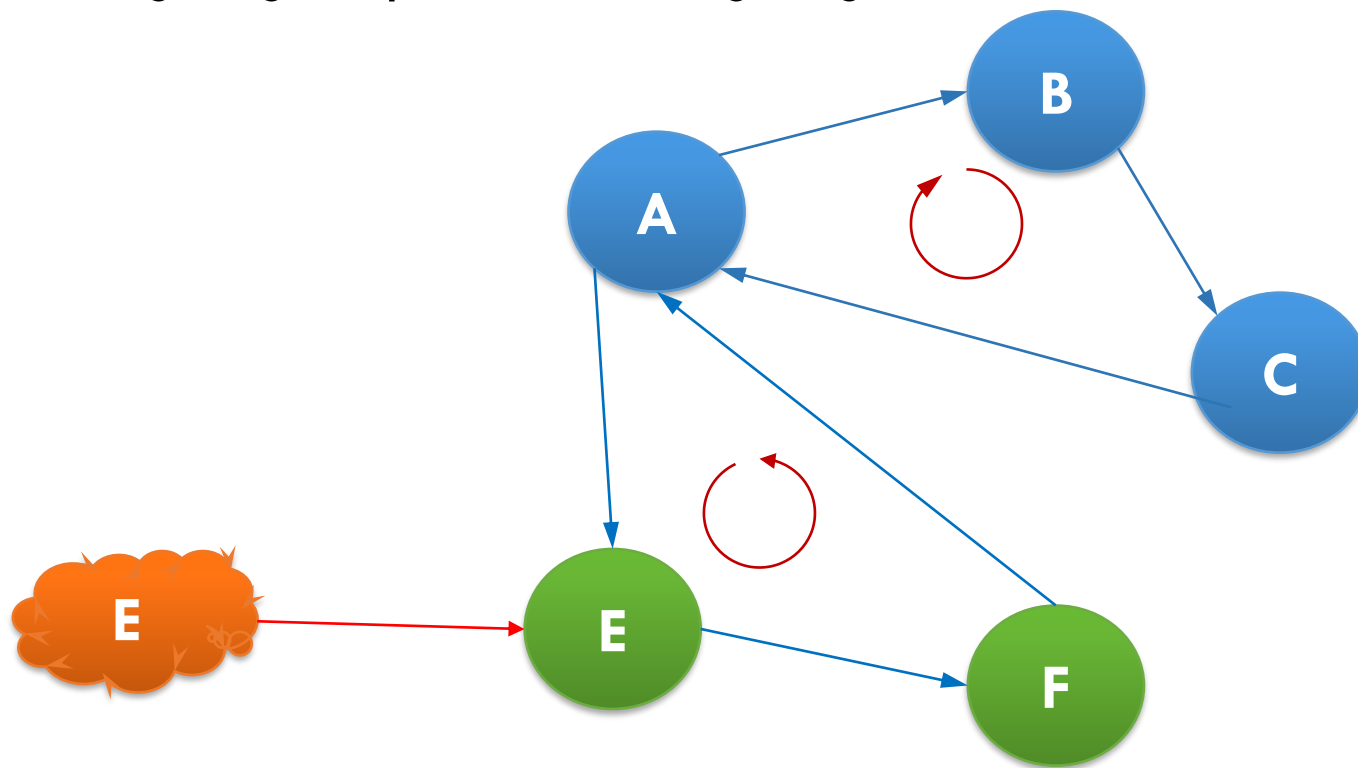
Sumber: Reisman, A. and Oral, M. (2003).
“Soft Systems Methodology: A Context
within a 50-year retrospective of OR/MS.”
Article in *Interfaces*, April 2005.



6. FEEDBACK LOOPS

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- **Feedback loop:** sistem dengan elemen yang saling mempengaruhi satu sama lain baik secara langsung maupun tidak langsung.



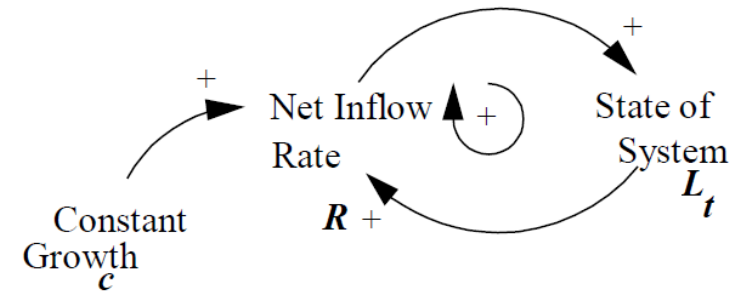


6. FEEDBACK LOOPS

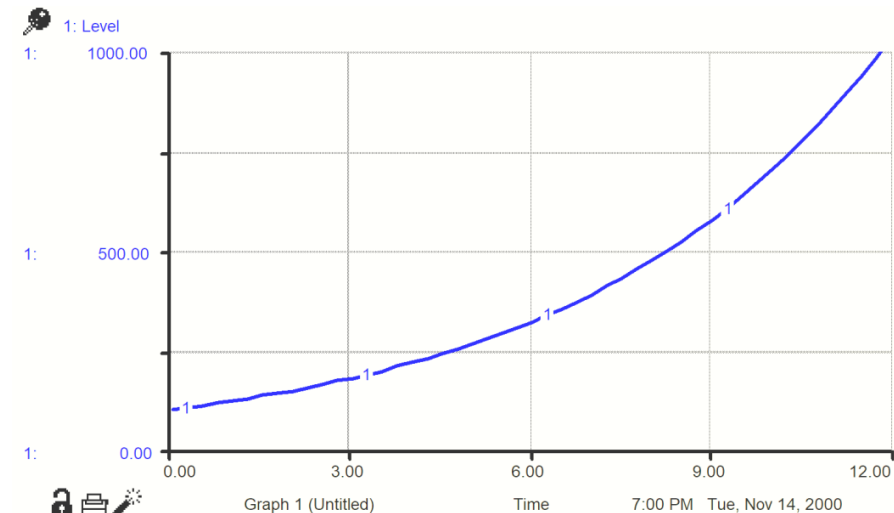
Negative and positive feedback loops:

- **Positive feedback loop:**
 - **Positive feedback** increases the discrepancy between the future state of the system and some reference state, such as an equilibrium state or a desired target state.
 - Positive feedback tends to **lead to instability**. The system either **explodes or kills itself**.

System structure



Behaviour of system



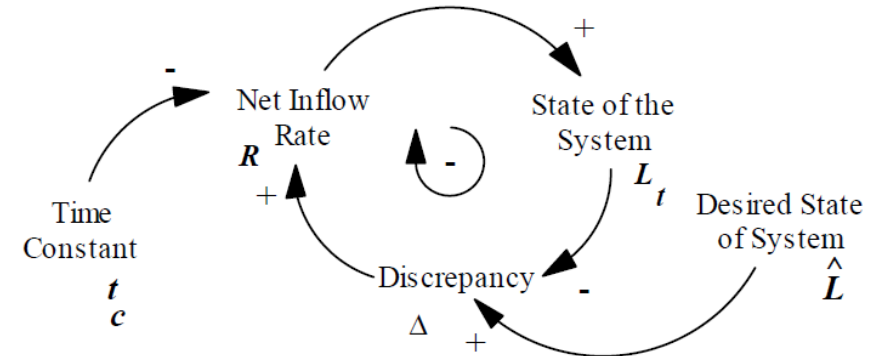


6. FEEDBACK LOOPS

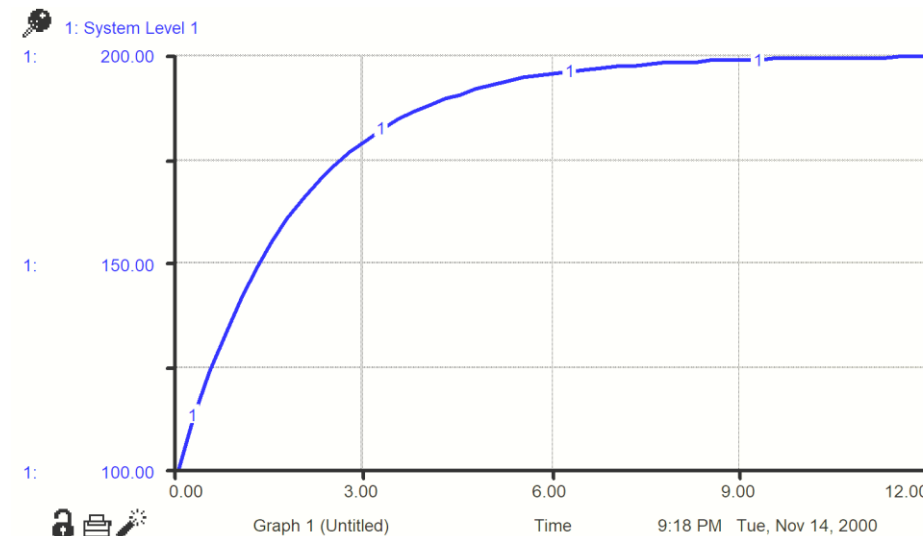
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- **Negative feedback loop:**
 - Negative feedback encourages the system to **achieve the system status at a certain condition (value)**.
 - It is therefore called as a goal seeking loop.

System structure



Behaviour of system output:





7. CONTROL OF SYSTEMS

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- Control of system is important to direct the system to behave in desired ways (pattern).
- Control is achieved by imposing something on the system in the form of inputs — a set of decisions, or decision rules, or simply an initial state for the system.
- Three conditions required to exercise control over system behaviour:
 1. **A target, objective, or goal** for the system to reach. For a deterministic system this may be a particular state of the system. For stochastic systems it may be a desirable steady state.
 2. **Capability of the system capable for reaching the target or goal.** Not all system has it, in particular the stochastic system. There is no way of guaranteeing a stochastic system to achieve the desired goal.
 3. **Some means of influencing system behaviour.** These are the control inputs (decisions, decision rules, or initial states). How these control inputs affect system behaviour is an important aspect of studying systems.



7. CONTROL OF SYSTEMS

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Types of control (System Theory):

1. Open loop or forward controls:

- Inputs imposed on the system based only on the prediction of how the system behaviour responds to them.
- No account is taken of how the system actually responds to the control inputs.
- Open loop controls are often in the form of a recipe or a set of rules to follow to ensure the desired condition is achieved.



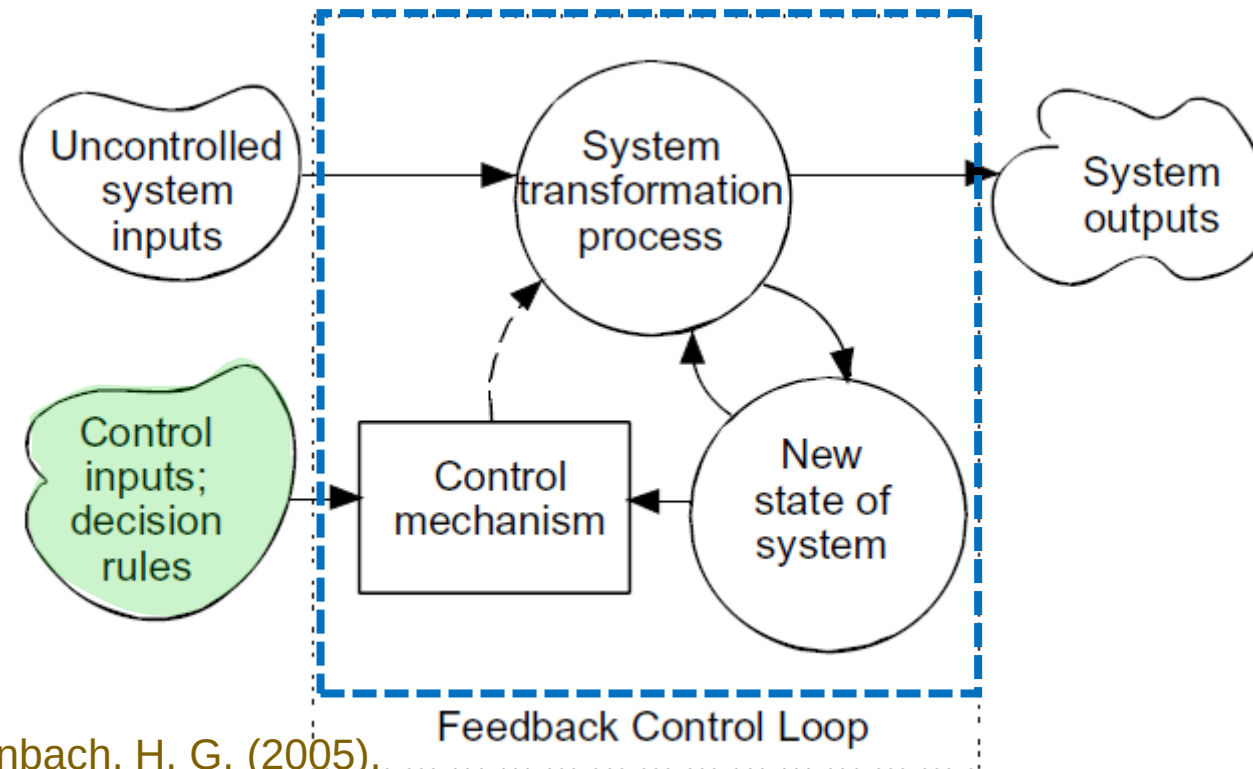


7. CONTROL OF SYSTEMS

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2. Closed loop or feedback controls:

- It has a mechanism to adjust the system to achieve the desired goal based on the gap between the desired goal and the actual achievement at the previous time period.



Adapted from Daellenbach, H. G. (2005),



7. CONTROL OF SYSTEMS

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2. Closed loop or feedback controls:

- Koreksi



7. CONTROL OF SYSTEMS

Key differences between feedback control and self-regulation control (system):

Aspect	Feedback Control	Self-Regulation Control
Control Mechanism	Requires external feedback and correction	Adjust itself through internal mechanism
Monitoring	Continuous monitoring and adjustment	Minimal or no external monitoring needed
Responsiveness	Quick responses to deviations	Slower, relies on system's natural balancing
Examples	Thermostat, cruise control in cars	Human body temperature regulation, ecosystem
Dependence	Depends on external controller and sensors	Autonomous, internal adjustment mechanisms

→ **Feedback control** requires **active monitoring and external correction**, while **self-regulation control** relies **on the system's built-in ability to maintain balance** without ongoing external intervention.

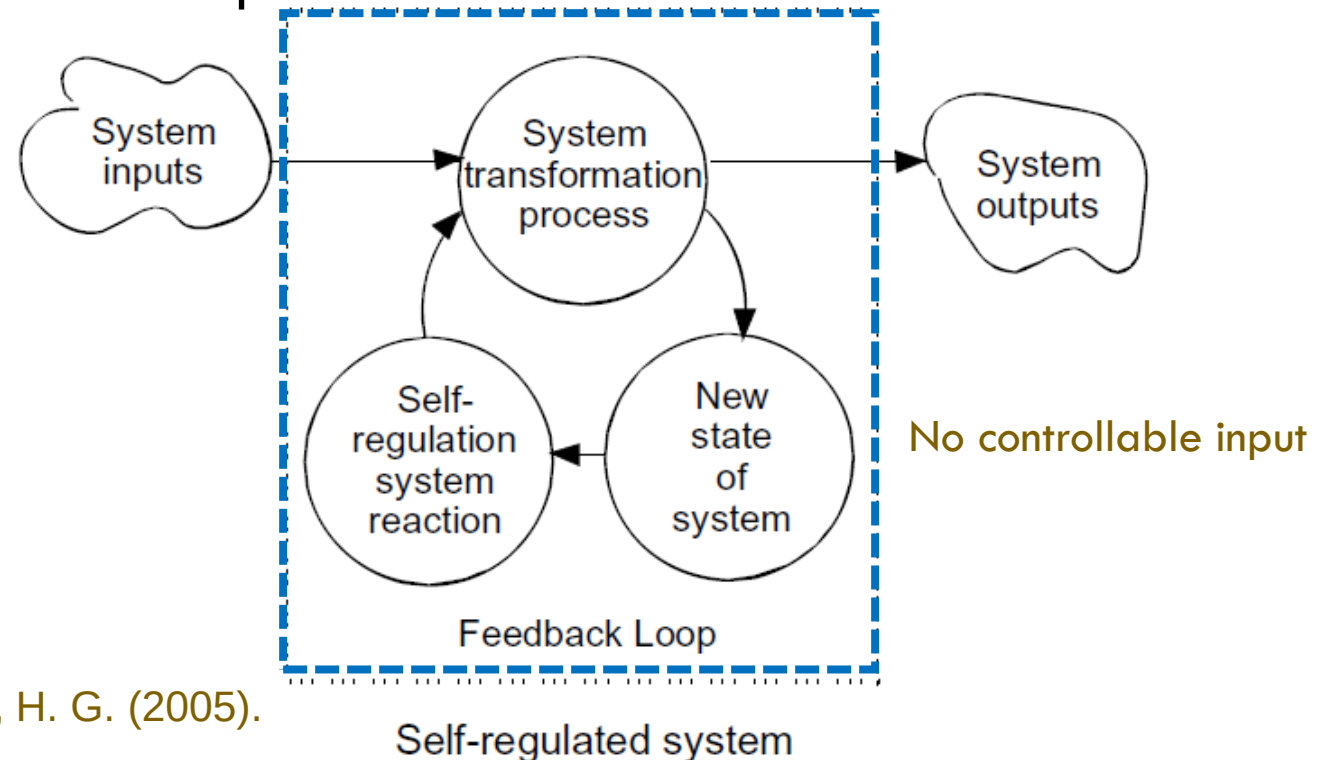


7. CONTROL OF SYSTEMS

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3. Self-regulation control (system):

- Feedback loops that occur in many natural systems, particularly biological and ecological systems, where they help regulate the behaviour of these systems.
- It helps the system to achieve its equilibrium state.



Adapted from Daellenbach, H. G. (2005).



7. CONTROL OF SYSTEMS

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4. Feed-forward control:

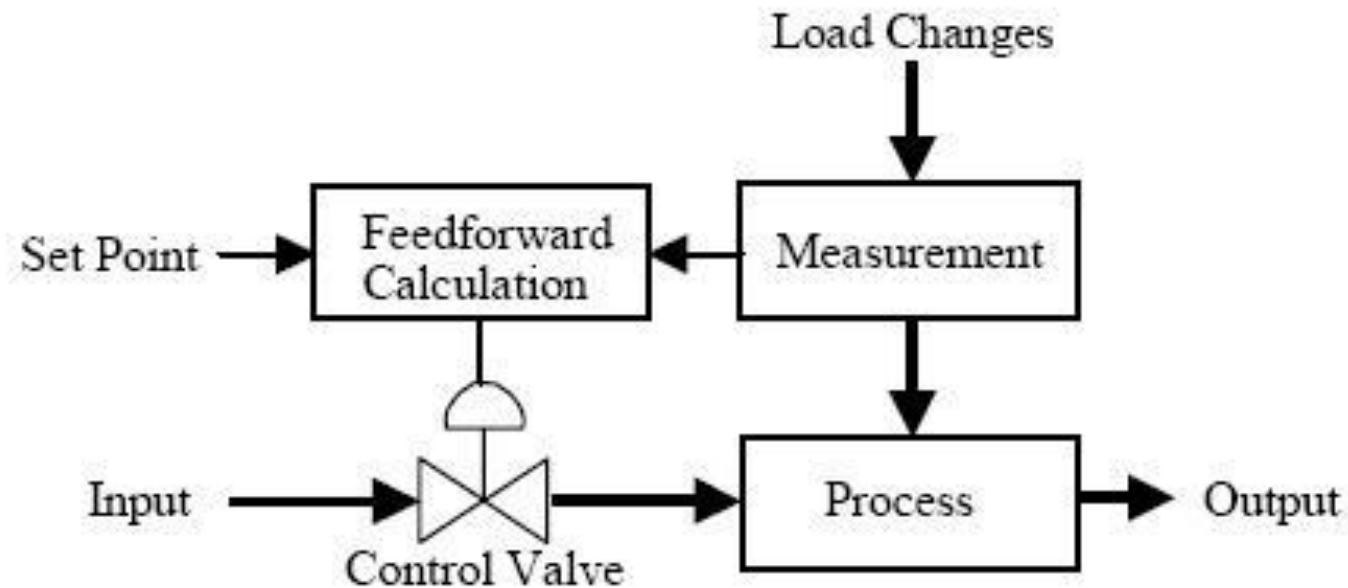
- Feed-forward control mechanism **predicts the impact of changes in input variables** (controllable and uncontrollable) on system behaviour (state variables).
- On the contrary, a **feedback control mechanism** reacts to changes in some critical state variables or outputs.
- **Examples:**
 - control of chemical processes
 - firm system to forecast future economic and demand trends and technological change in order to take advantage of growth opportunities or avoid potential disasters due to an economic downturn or a change in technology.



7. CONTROL OF SYSTEMS

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Feed-forward control:



Source: <https://www.wakeindustrial.com/site-blog/feedback-feedforward-control-systems>



8. SYSTEM STRUCTURE AND BEHAVIOURS

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Response Lags In Systems (time delay)

- Respon of a system to an adjustment may not be instantanneous. There is time **delay between adjustment and system response**.
- Time delay affects the system behaviour.
- Time delay is one of building block that influences system behaviours.



8. SYSTEM STRUCTURE AND BEHAVIOURS

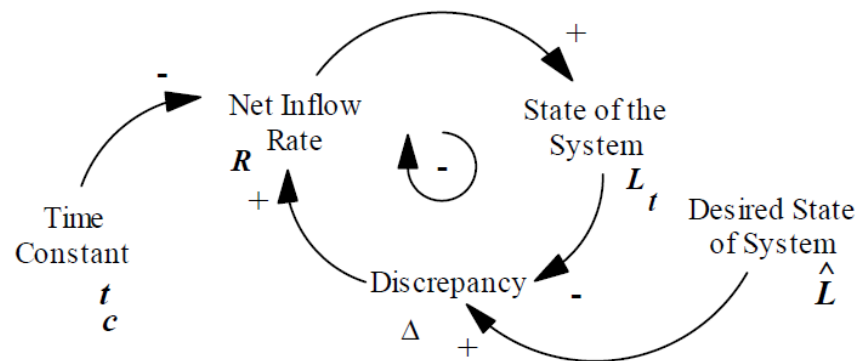
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Building blocks of system archetypes:

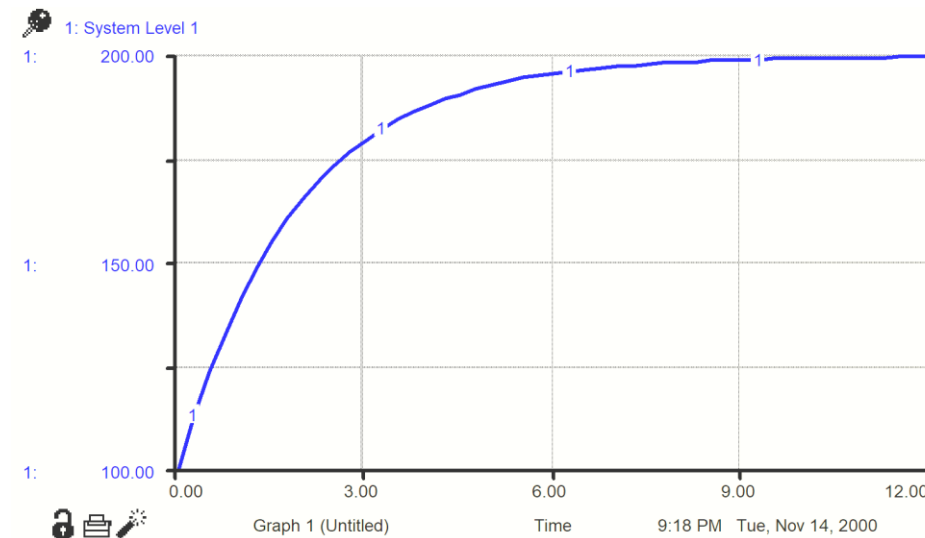
- Reinforcing processes
- Balancing processes
- Delays

Building block 1: Balancing process

System structure



Behaviour of system output:





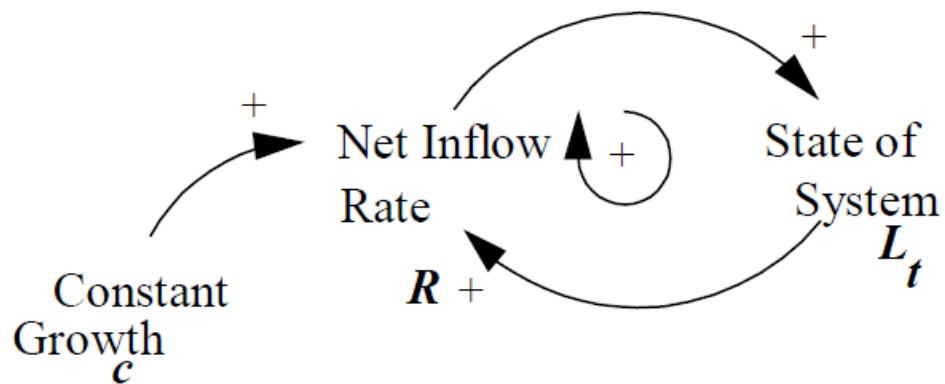
8. SYSTEM STRUCTURE AND BEHAVIOURS

Building block 2: Reinforcing process

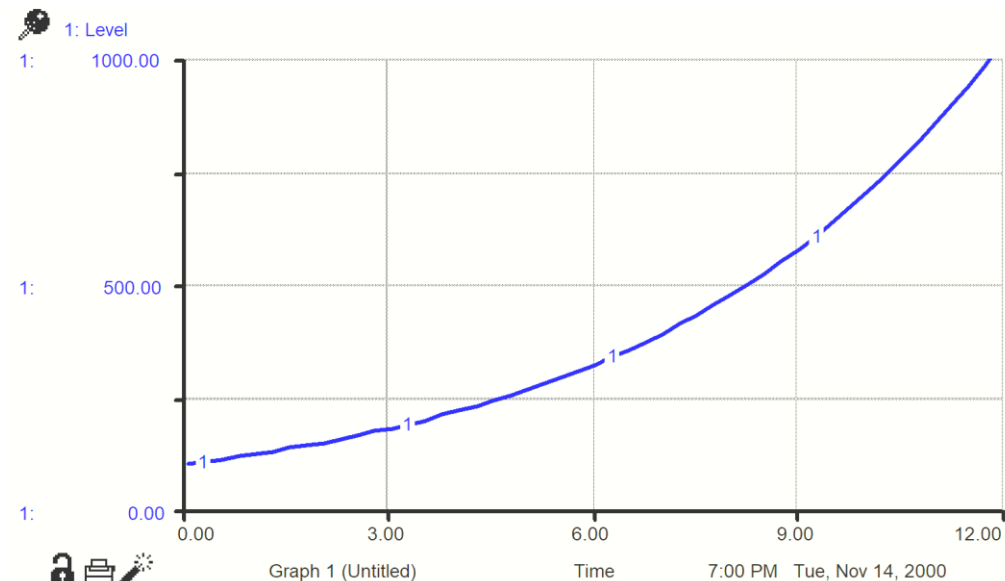
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- It creates exponential growth, amplifies deviations, and reinforces change.
- It consists of dominantly positive feedback loops.

System structure



Behaviour of system output:





8. SYSTEM STRUCTURE AND BEHAVIOURS

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Building block 3: Delay

- Time difference between the beginning of flow going into a conversion process and the beginning of flow going out from the process.
- It may occur in any flow, material, energy and information.
- An important factor that influences the dynamics of system behaviours.
- Delay creates transient response from the system.

Impact of delay on system behavior:

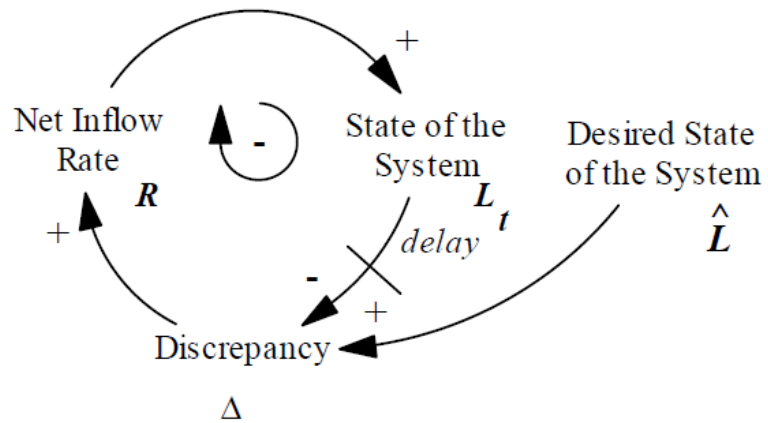
- Time delays in lead to the possibility that the state of the system will overshoot and oscillate around the carrying capacity.



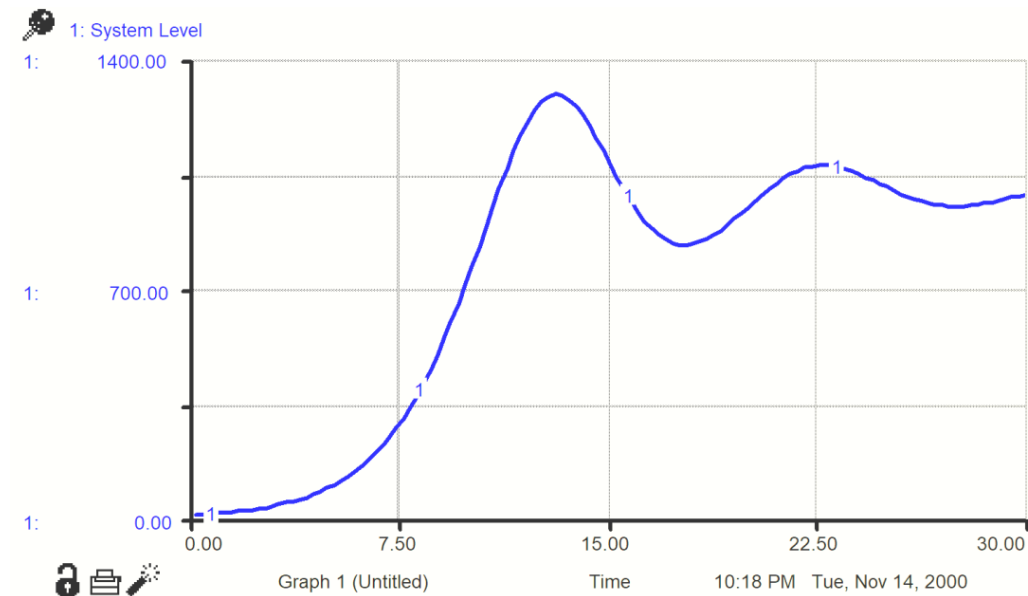
8. SYSTEM STRUCTURE AND BEHAVIOURS

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System structure



Behaviour of system output:





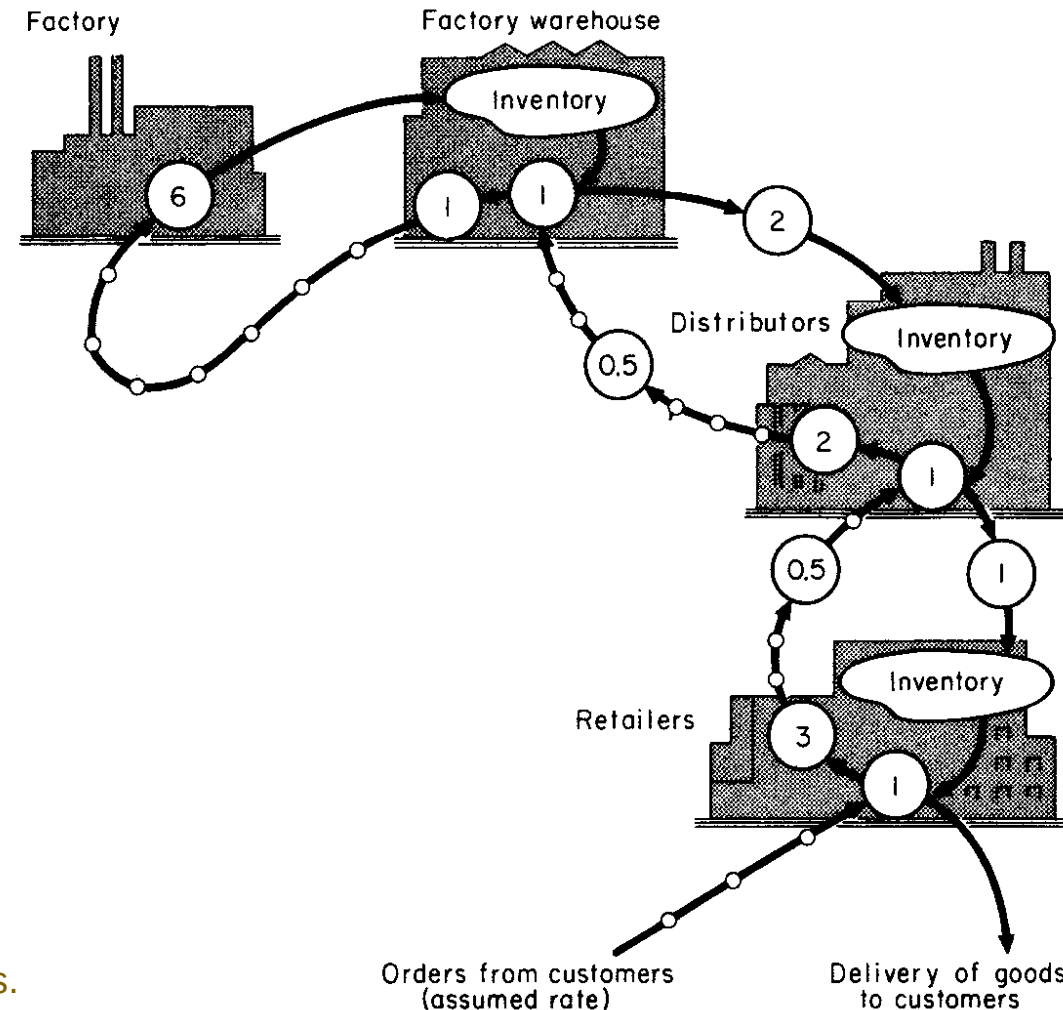
8. SYSTEM STRUCTURE AND BEHAVIOURS

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Ilustrasi Perilaku Sistem Rantai Pasok 3 Eselon

Keterangan:

- Aliran produk
- Aliran informasi



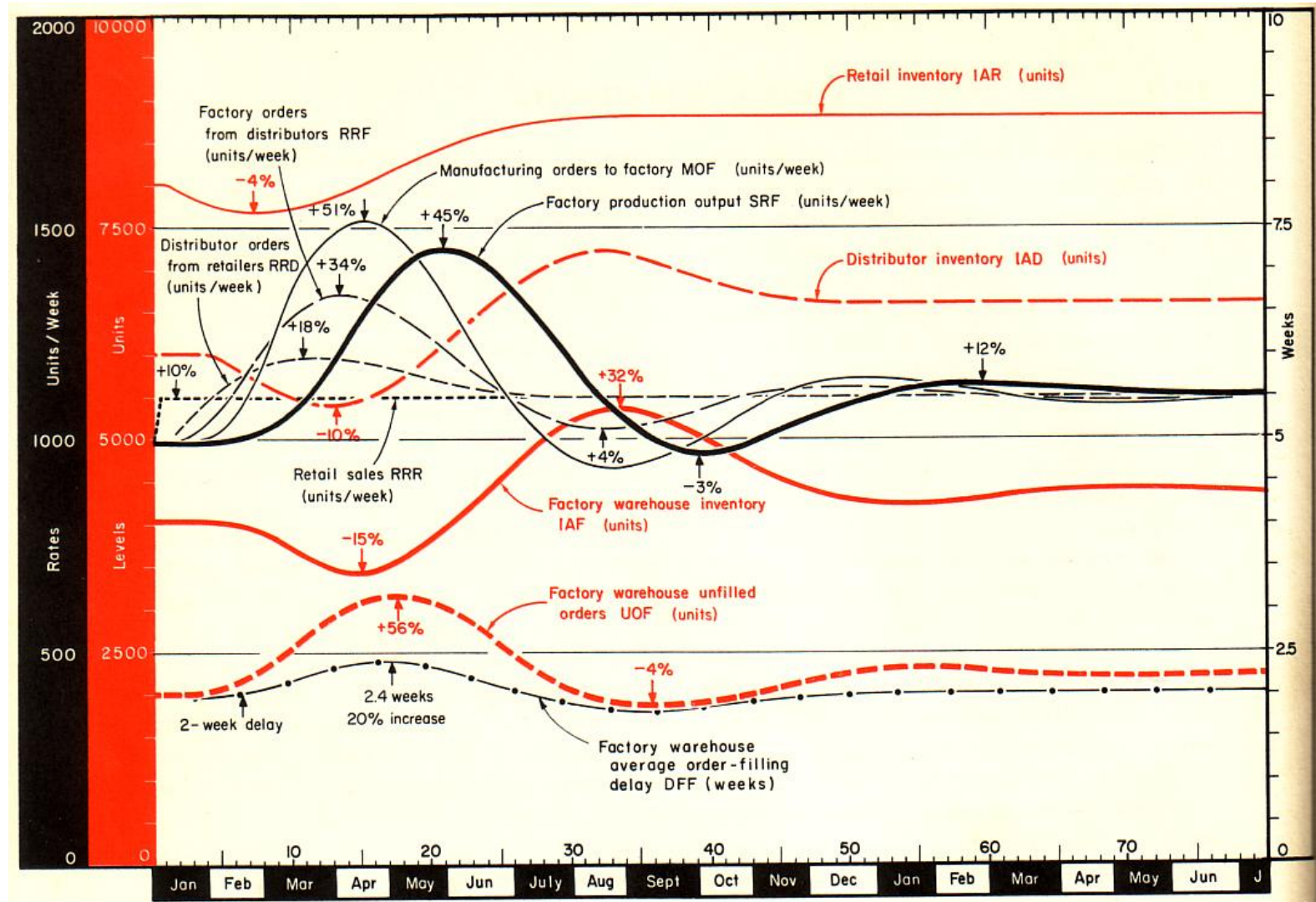
Source: Forrester (1961). Industrial Dynamics.



8. SYSTEM STRUCTURE AND BEHAVIOURS

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■ Respon Rantai Suplai thdp. Peningkatan 10% Demand:



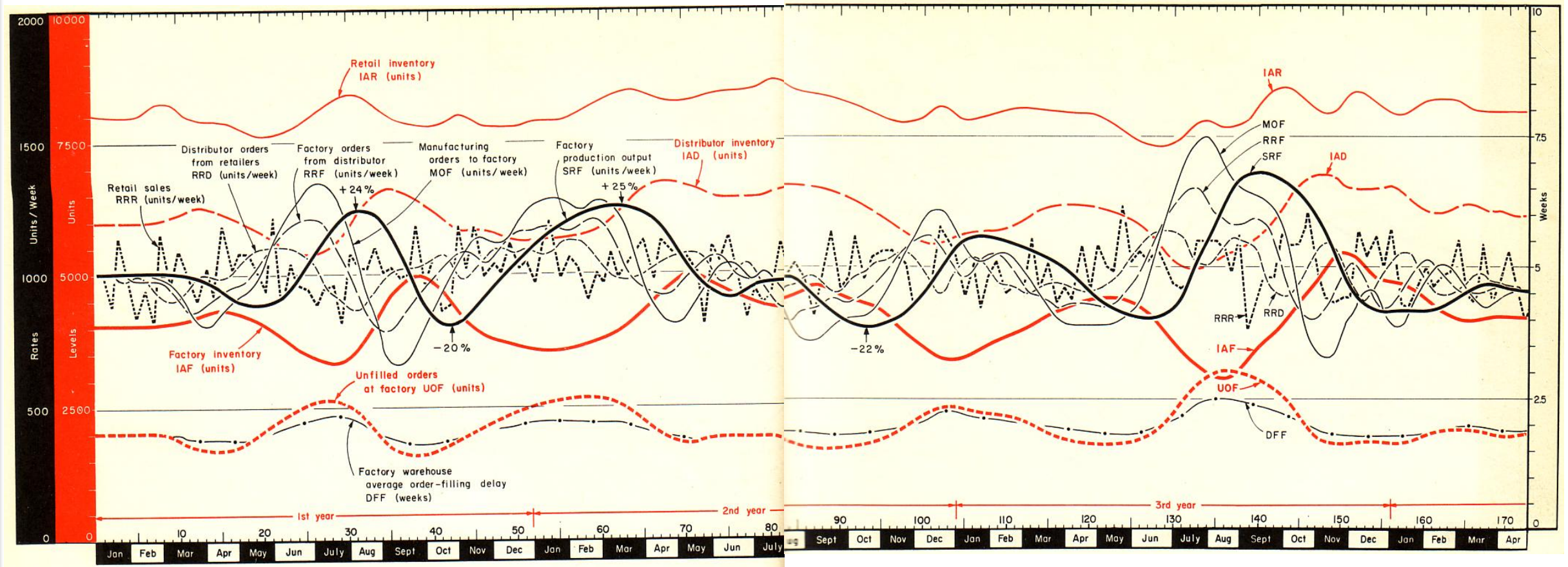
Source: Forrester (1961).
Industrial Dynamics.



8. SYSTEM STRUCTURE AND BEHAVIOURS

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■ Respon Rantai Suplai thdp. Deviasi Random Demand:



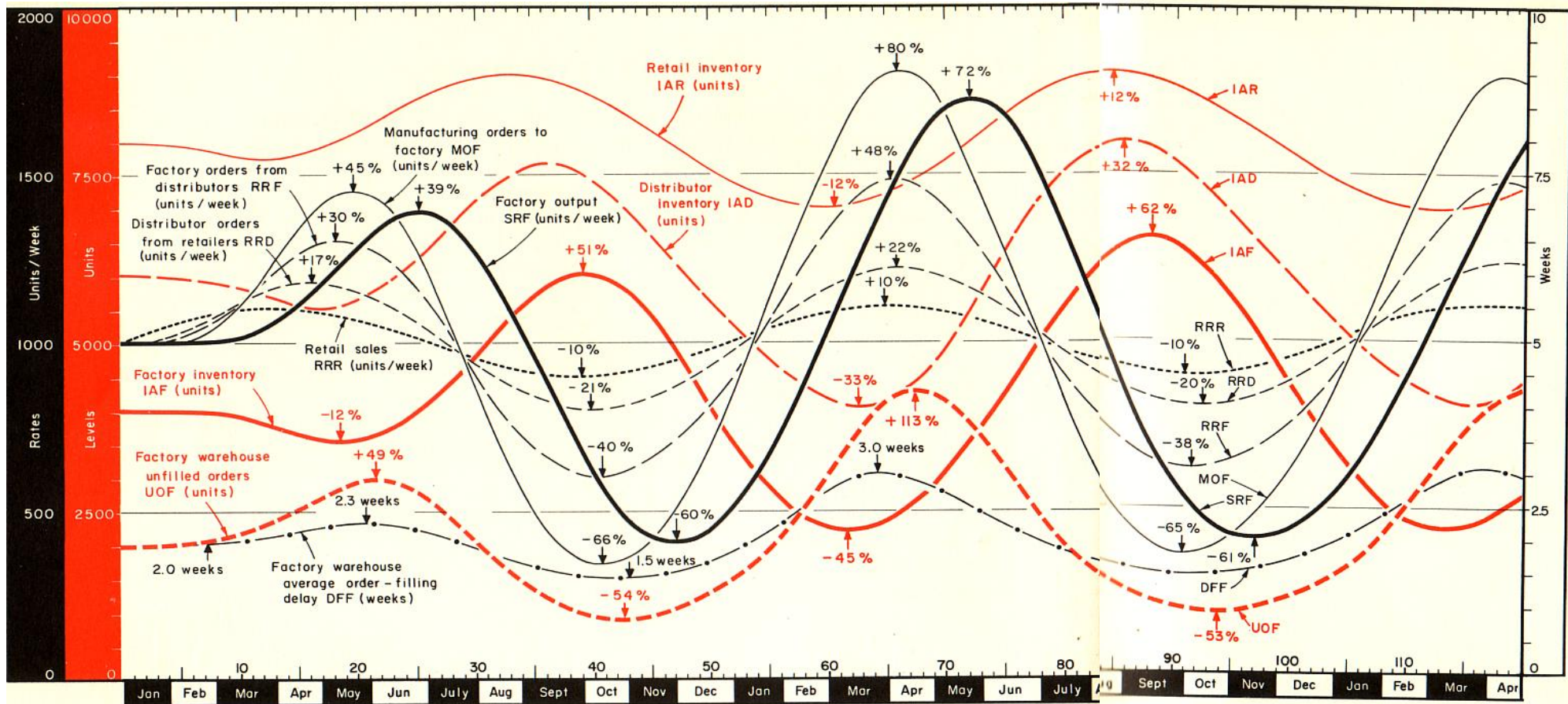
Source: Forrester (1961). Industrial Dynamics.



8. SYSTEM STRUCTURE AND BEHAVIOURS

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■ Respon Rantai Suplai thdp. Peningkatan & Penurunan 10% Demand:



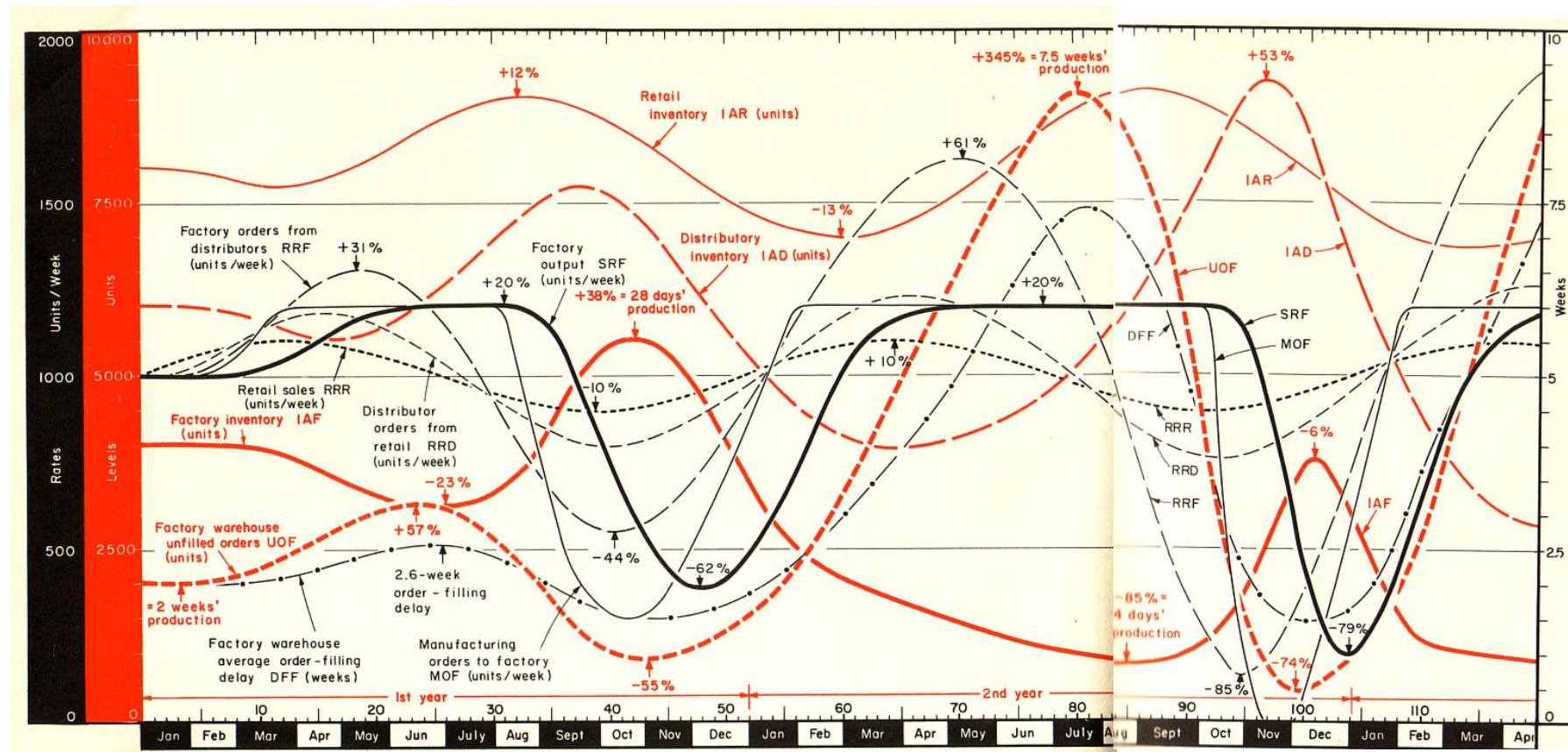
Source: Forrester (1961). Industrial Dynamics.



8. SYSTEM STRUCTURE AND BEHAVIOURS

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■ Dampak Fluktuasi Demand terhadap Pabrik (kapasitas lebih 20% dari rata-rata demand)





8. SYSTEM STRUCTURE AND BEHAVIOURS

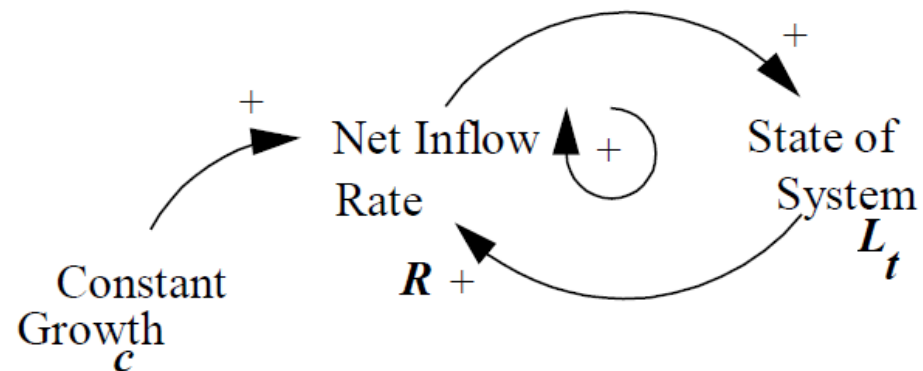
System Archetypes

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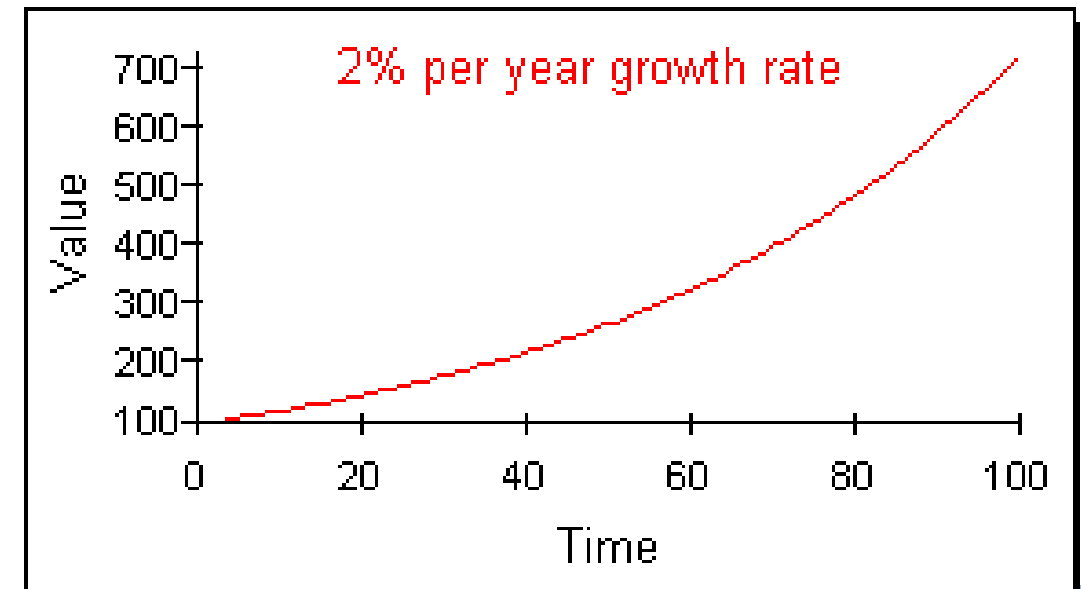
a. Exponential Growth

Perilaku yang dihasilkan dari loop kausal dengan polaritas POSITIF. Loop ini membuat *growth*, *amplify deviations*, and *reinforce change*.

Struktur sistem:



Exponential Family of Time Paths





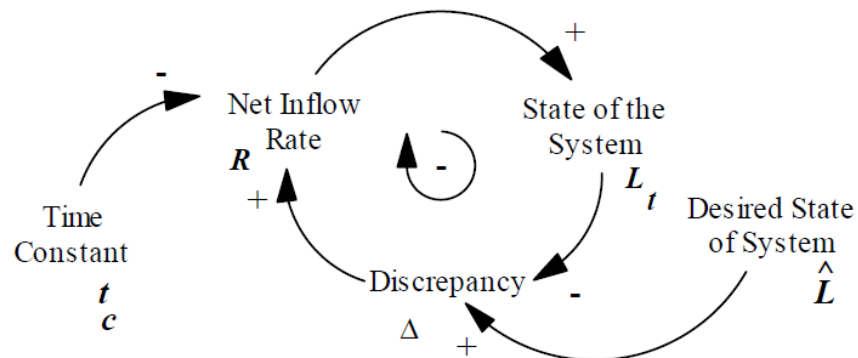
8. SYSTEM STRUCTURE AND BEHAVIOURS

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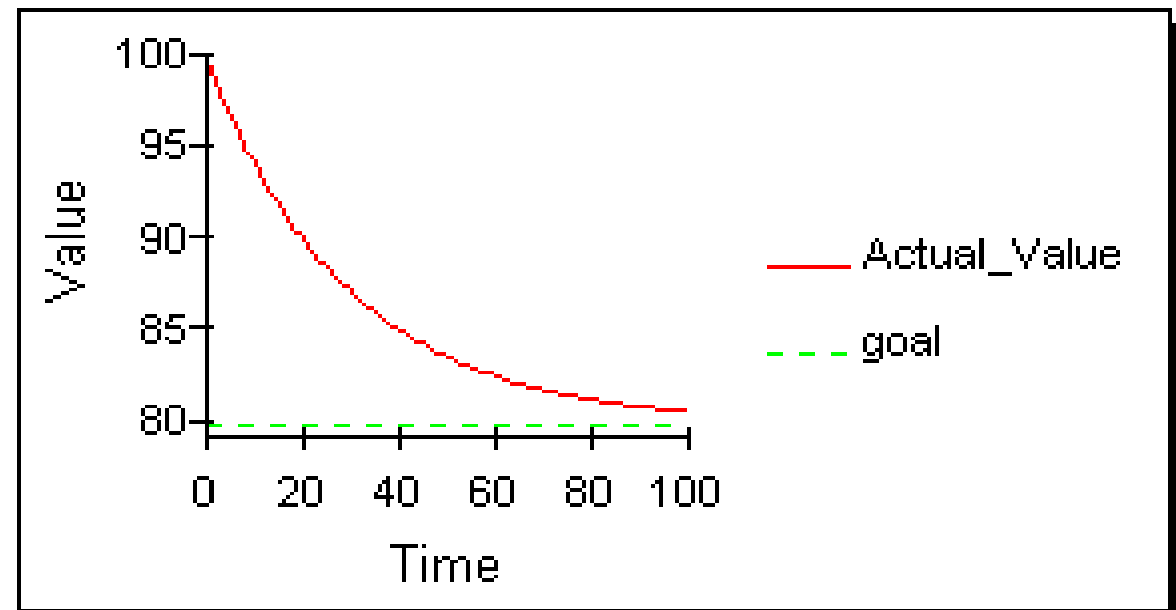
b. Goal Seeking

Perilaku yang dihasilkan dari loop kausal dengan polaritas NEGATIF. Loop ini mencari *balance, equilibrium, and stasis*.

Struktur sistem:



Goal Seeking Family of Time Paths





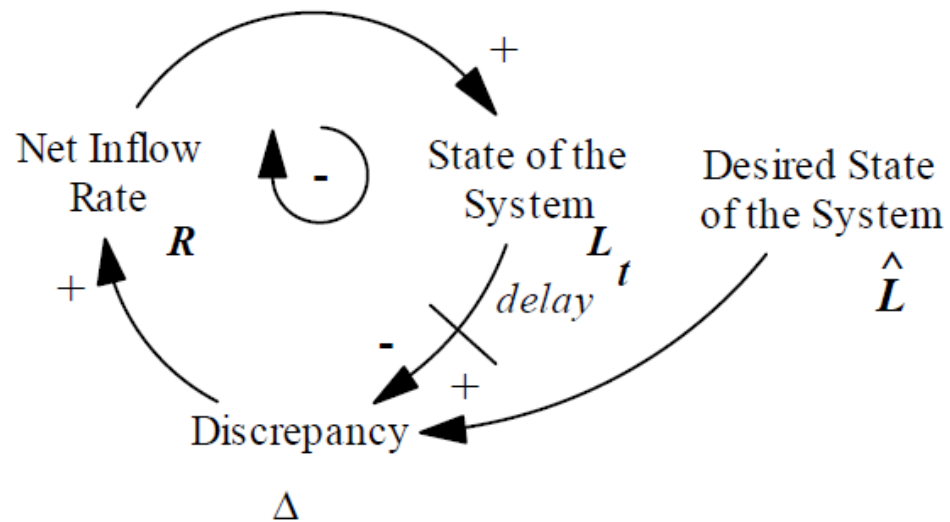
8. SYSTEM STRUCTURE AND BEHAVIOURS

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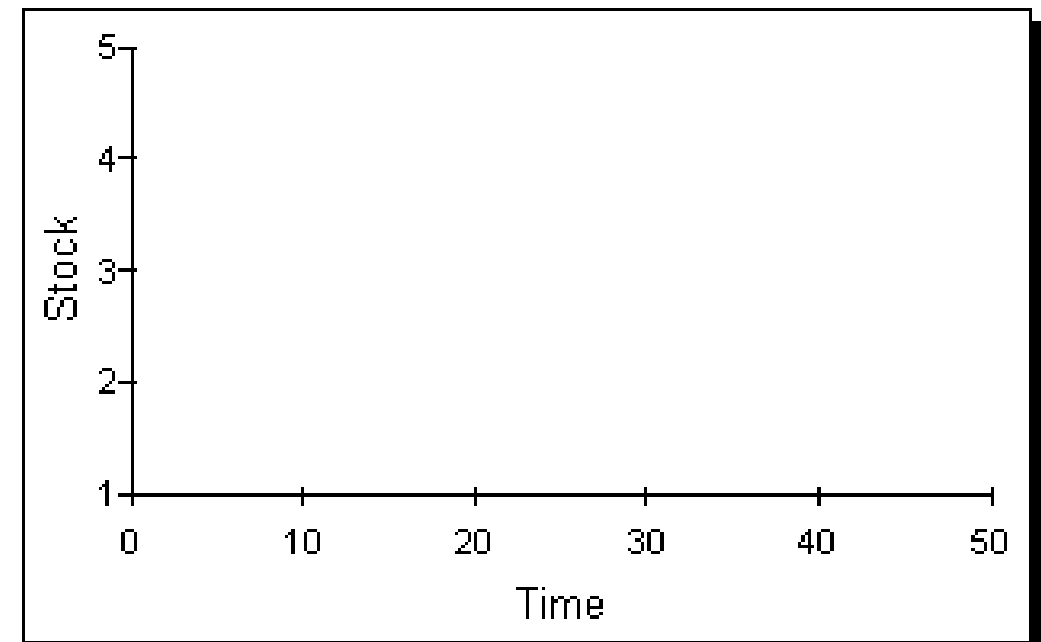
c. Oscillation

Perilaku yang disebabkan oleh adanya **keterlambatan waktu** (*time delay*) dalam merespon status sistem. *Time delays* menyebabkan keadaan sistem secara konstan melebihi sasaran atau kondisi ekuilibrium, atau sebaliknya lebih rendah, dan seterusnya → status sistem berosilasi di sekitar sasaran.

Struktur Sistem:



Oscillation Family of Time Paths





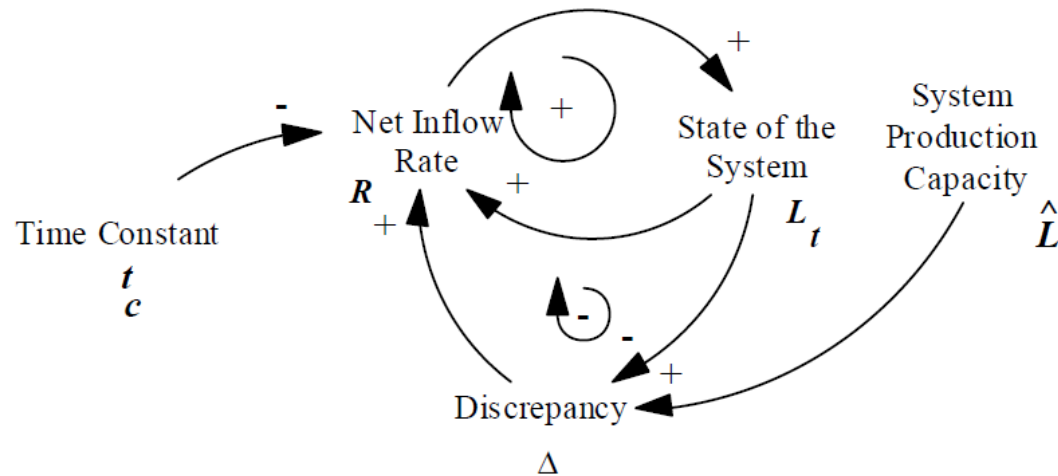
8. SYSTEM STRUCTURE AND BEHAVIOURS

d. S-shaped Growth

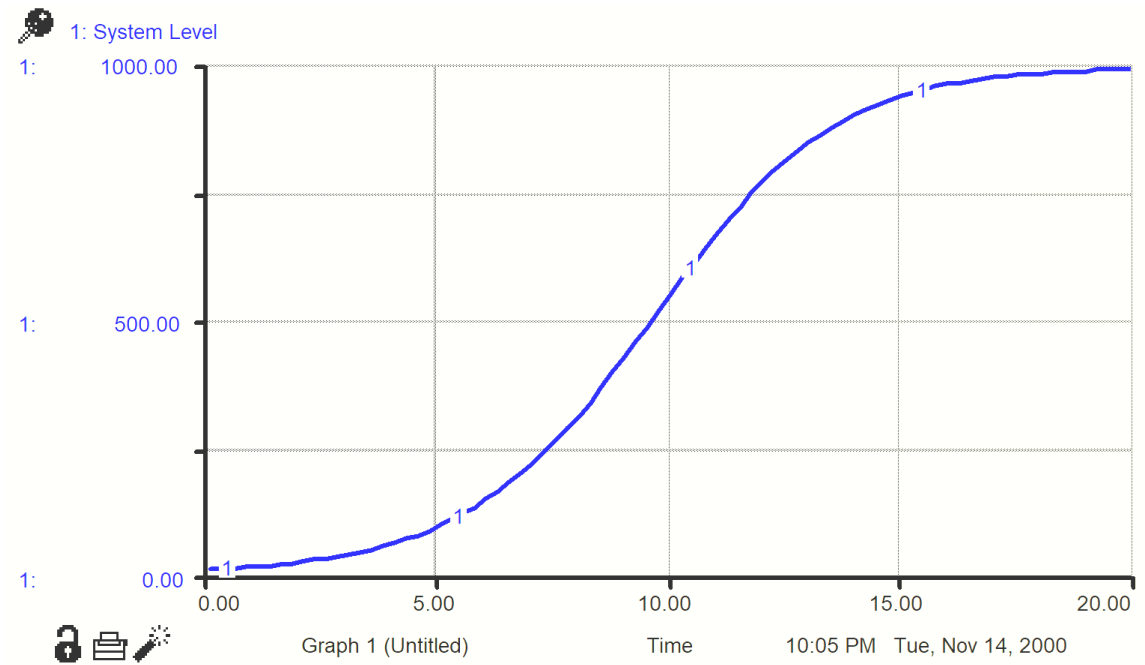
Perilaku yang disebabkan adanya keterbatasan sumberdaya yang diperlukan untuk pertumbuhan.

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Struktur Sistem:



S-shaped Growth of Time Paths





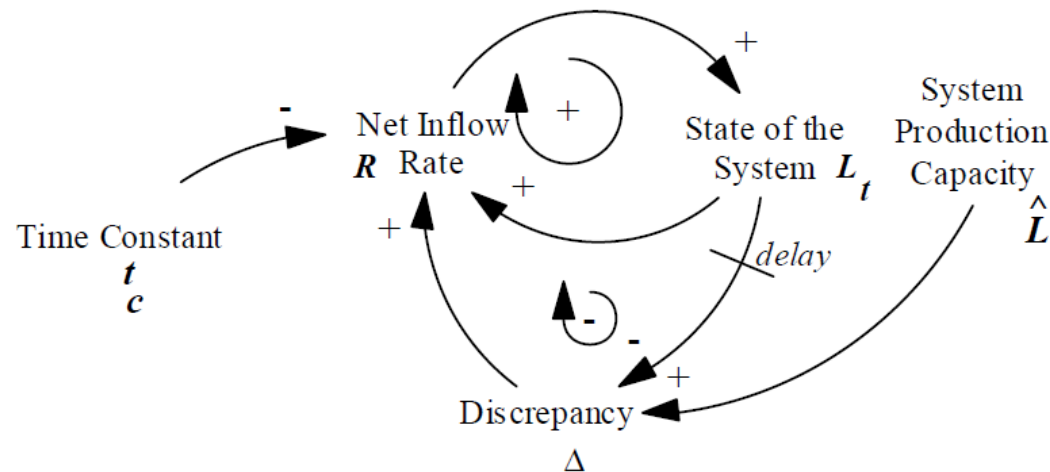
8. SYSTEM STRUCTURE AND BEHAVIOURS

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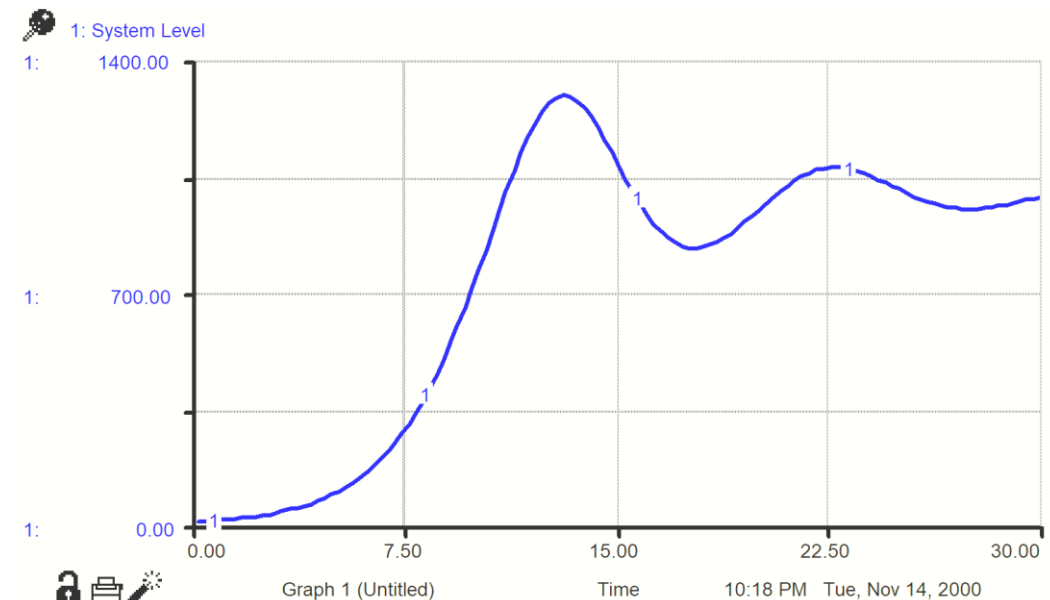
e. S-shaped Growth with Overshoot

Time **delays** dalam loop kausal dengan polaritas negatif menyebabkan kemungkinan status sistem akan *overshoot* dan berosilasi di sekitar kapasitas sumber daya yang diperlukan.

Struktur Sistem:



S-shaped Growth and Overshoot of Time Paths





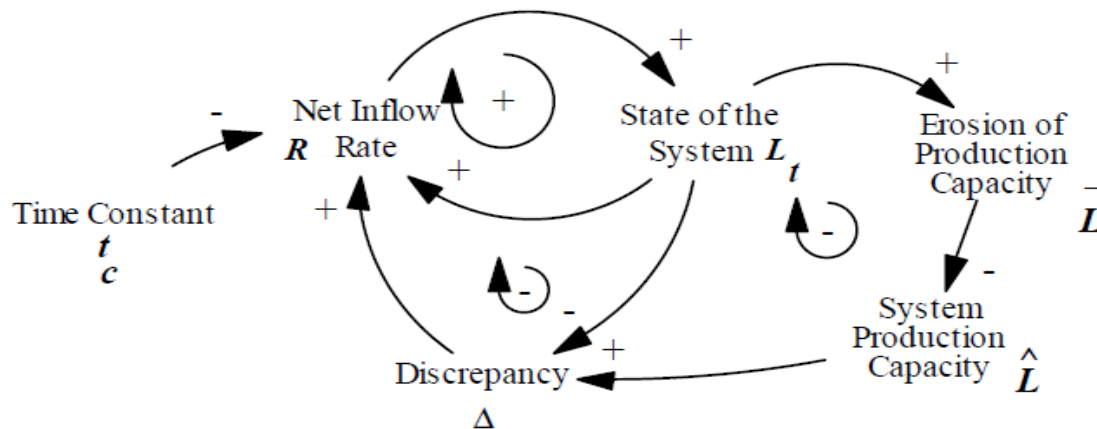
8. SYSTEM STRUCTURE AND BEHAVIOURS

f. Overshoot and Collapse

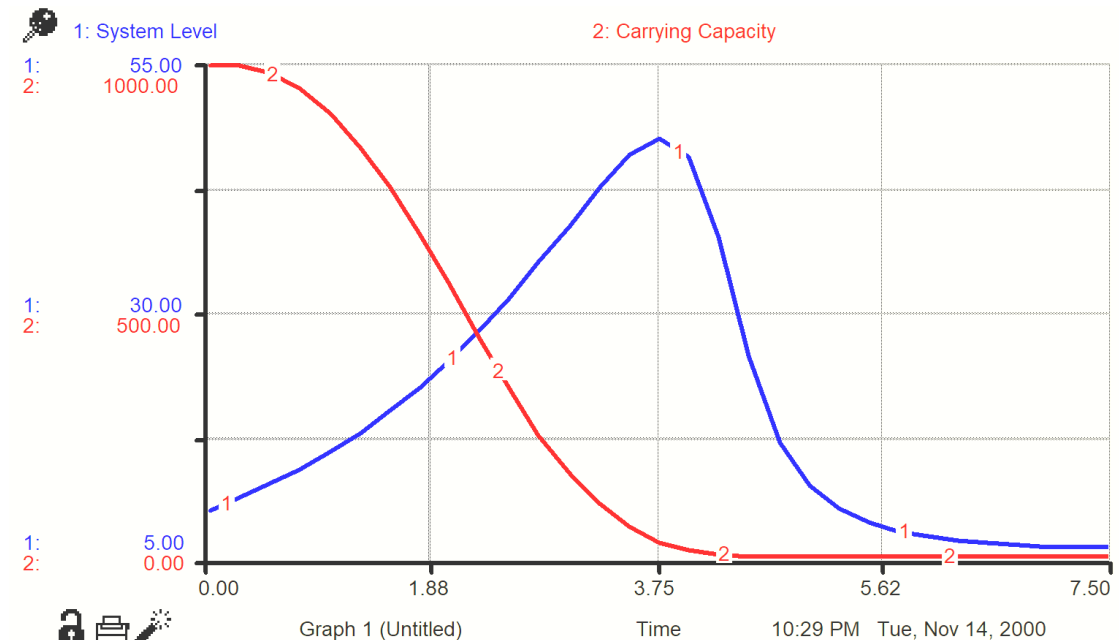
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Perilaku ini disebabkan oleh daya dukung sumber daya yang diperlukan untuk pertumbuhan menurun dan habis.

Struktur Sistem:



Overshoot and Collapse of Time Paths



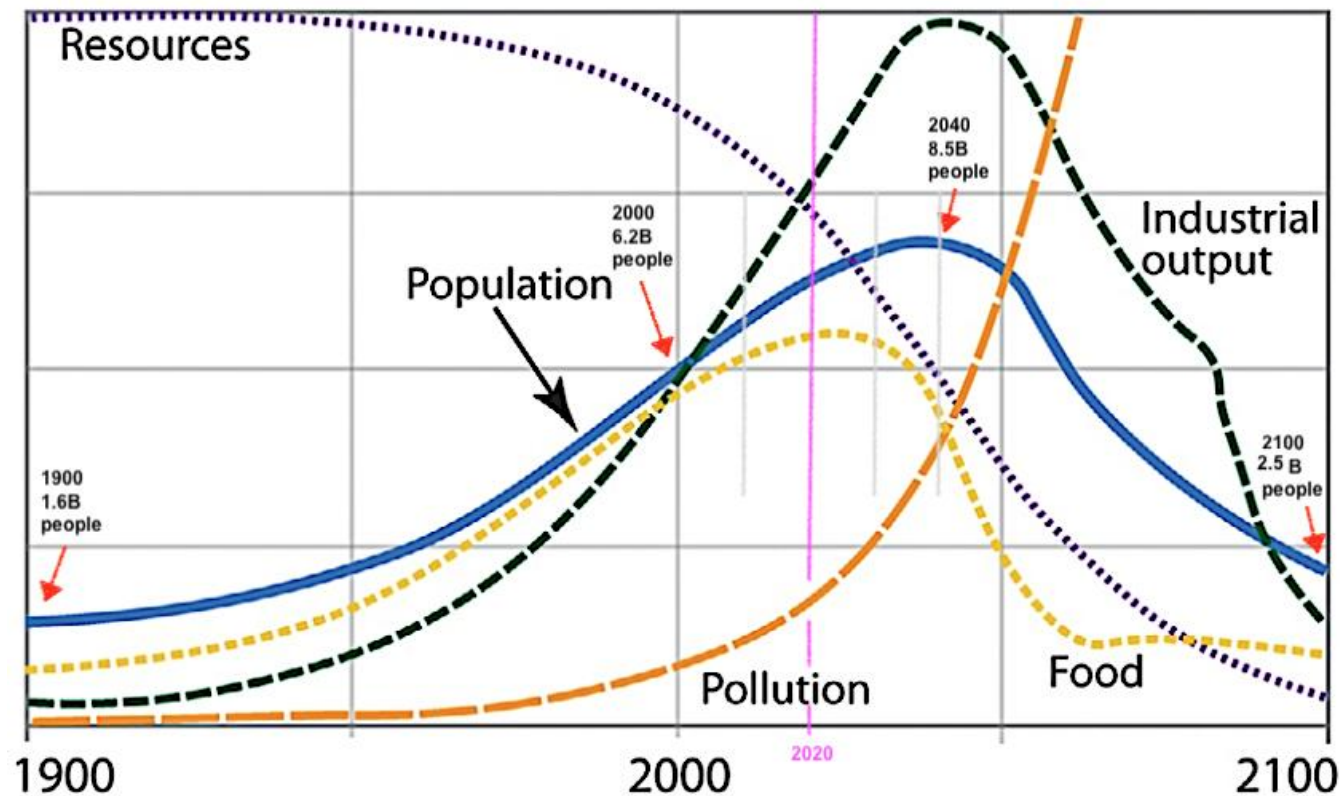


8. SYSTEM STRUCTURE AND BEHAVIOURS

Limits to Growth

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MIT study funded by a group of European industrialists calling themselves the Club of Rome



Source:
https://www.joboneforhumanity.org/what_is_the_the_club

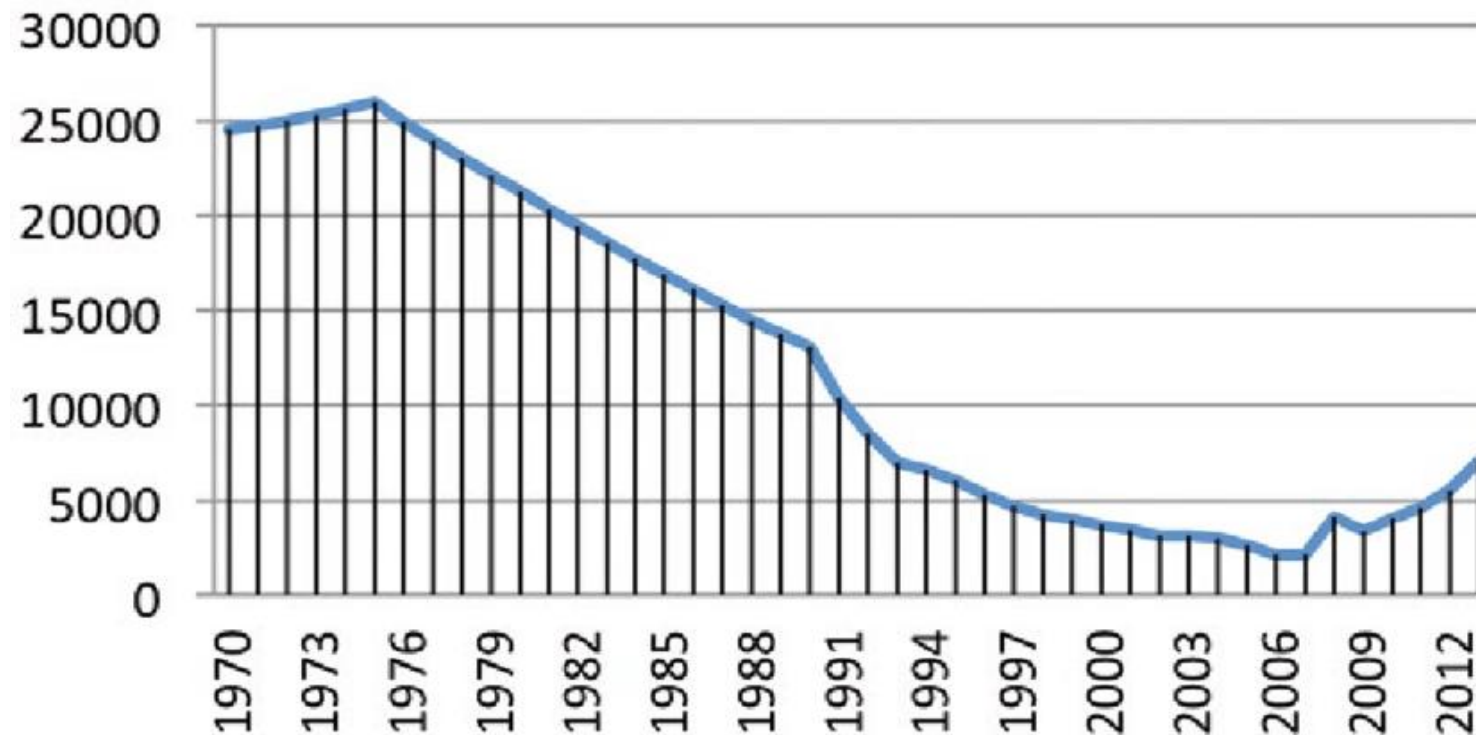


8. SYSTEM STRUCTURE AND BEHAVIOURS

The Story of Nauru

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GDP per Capita of Nauru (US\$ per cap)



Brut.





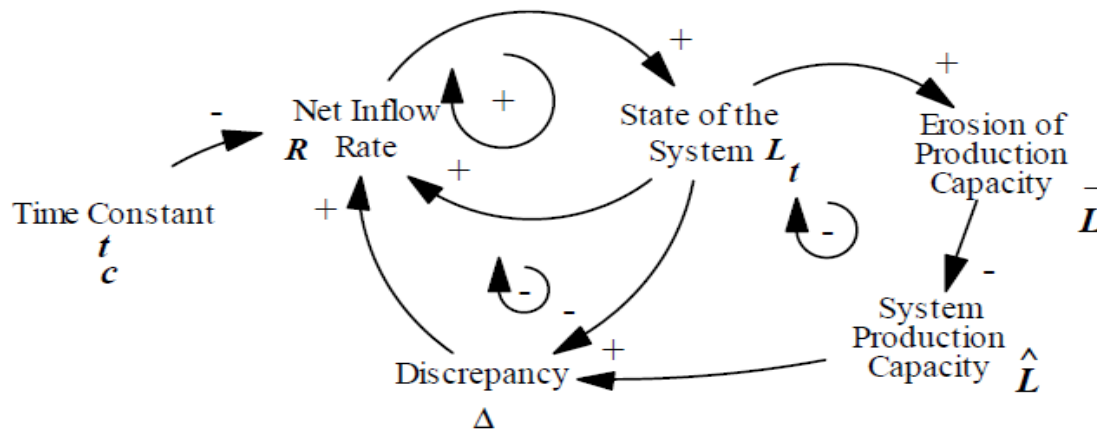
8. SYSTEM STRUCTURE AND BEHAVIOURS

f. Overshoot and Collapse

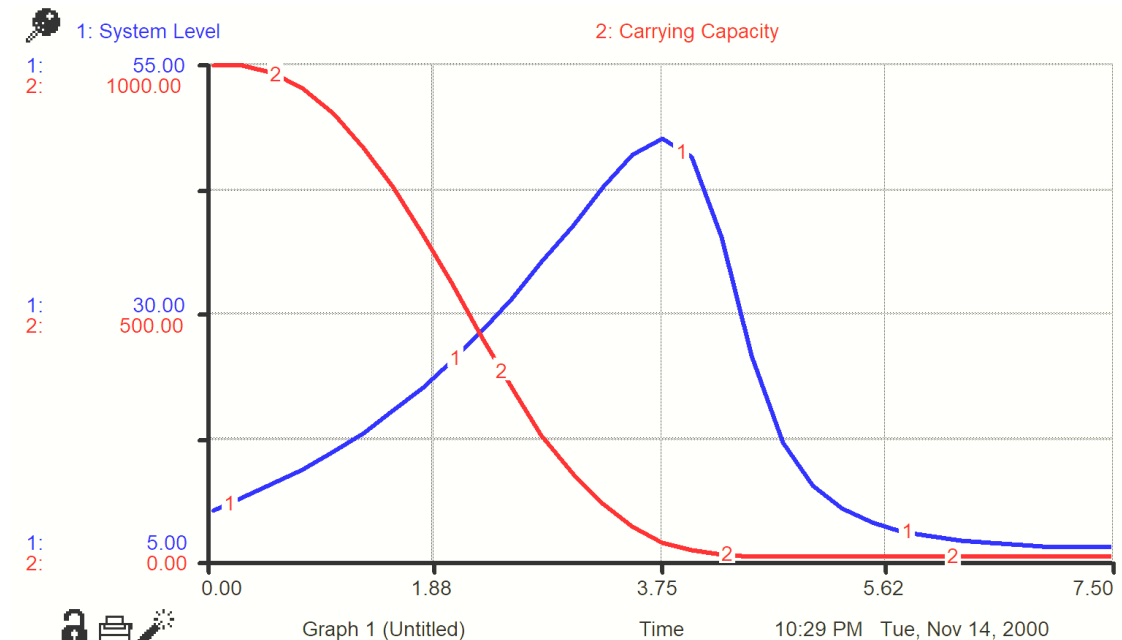
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Perilaku ini disebabkan oleh daya dukung sumber daya yang diperlukan untuk pertumbuhan menurun dan habis.

Struktur Sistem:



Overshoot and Collapse of Time Paths

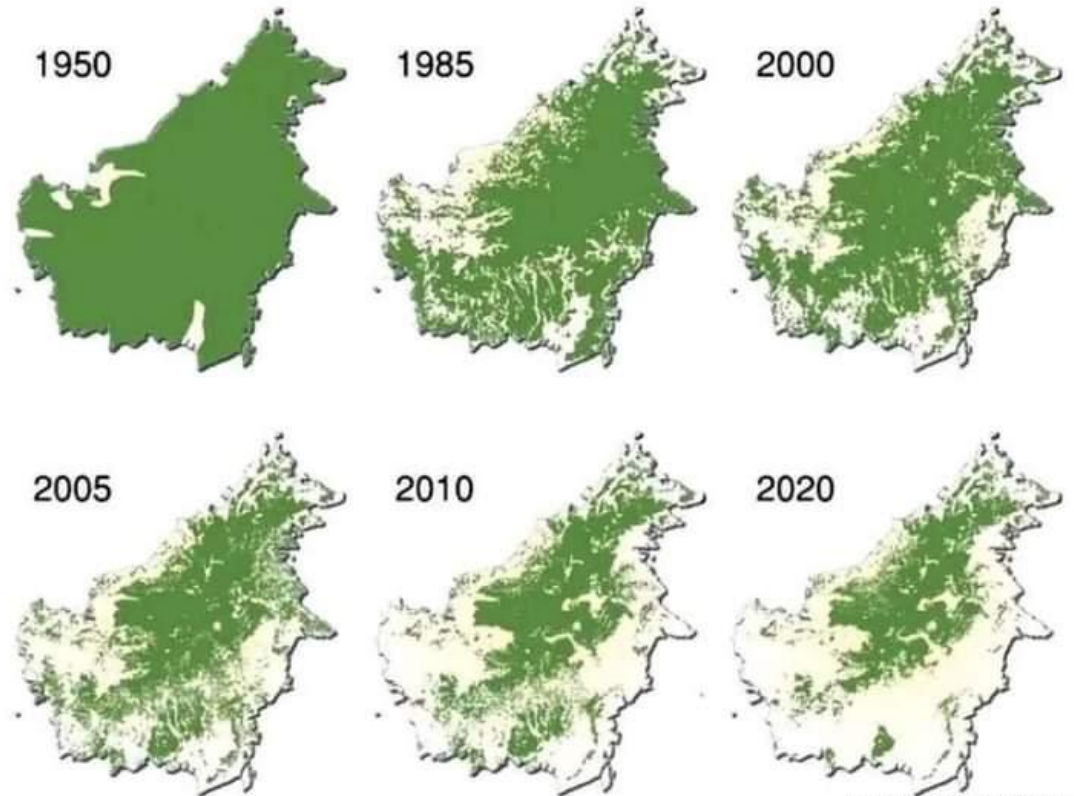




8. SYSTEM STRUCTURE AND BEHAVIOURS

f. Overshoot and Collapse

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**FOREST COVER IN
BORNEO FROM 1950-2020**



PHOTO CREDIT FROM THE VARIOUS SOURCES

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